Experiment 06

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

Lab Outcome: Mapped LO1

LO1: Apply Agile methodologies and use Jira for Sprint Planning, task tracking and integrating with DevOps workflow.

Theory:

What is Jira?

Jira is an Agile project management and issue tracking tool developed by Atlassian. It helps software engineering teams plan sprints, track tasks, manage bugs, and monitor releases across customizable workflows.

Core Concepts & Hierarchy

- Project: A shared workspace representing a product, application, or team initiative.
- Issue Hierarchy:
  - Epic: A large objective or high-level feature broken down into smaller tasks (e.g., Core Infrastructure).
  - Story: A end-user feature written from a user perspective (e.g., Storefront UI).
  - Task / Bug: Technical implementations or defects to resolve.
  - Sub-task: Granular steps within a story or task.
- Product Backlog: A prioritized list of all pending features, enhancements, and bug fixes awaiting assignment.
- Sprint & Scrum Board: A time-boxed cycle (typically 1–2 weeks) where team members pull items from the backlog into active workflow columns (To Do, In Progress, In Review, Done).
- Story Points: Relative estimates assessing effort, technical complexity, and risk for each work item.

Jira Setup & Deployment Models

- Jira Cloud (SaaS): Hosted directly by Atlassian. No manual server maintenance or database installations required; teams create an organization site (e.g., company.atlassian.net) and get instant access via a web browser.
- Jira Data Center (Self-Managed): Installed on private infrastructure or dedicated cloud servers (AWS/Azure) for enterprises requiring localized data governance, custom security policies, and high-availability clustering.

DevOps Integration & Automation Lifecycle

- Version Control Linkage: Integrates with code repositories (such as GitHub, GitLab, or Bitbucket) through official marketplace apps (e.g., GitHub for Jira).
- Smart Issue Keys: Mentioning ticket identifiers (e.g., SCRUM-4, SCRUM-6) in branch names, commit messages, or Pull Request titles binds VCS activity directly to the Jira issue panel.

• Jira Automation Rules: Event-driven rules that trigger state transitions across the board (e.g., automatically shifting a ticket to In Review when a Pull Request is opened, or transitioning to Done upon a merged PR).

Team Collaboration & Access Management

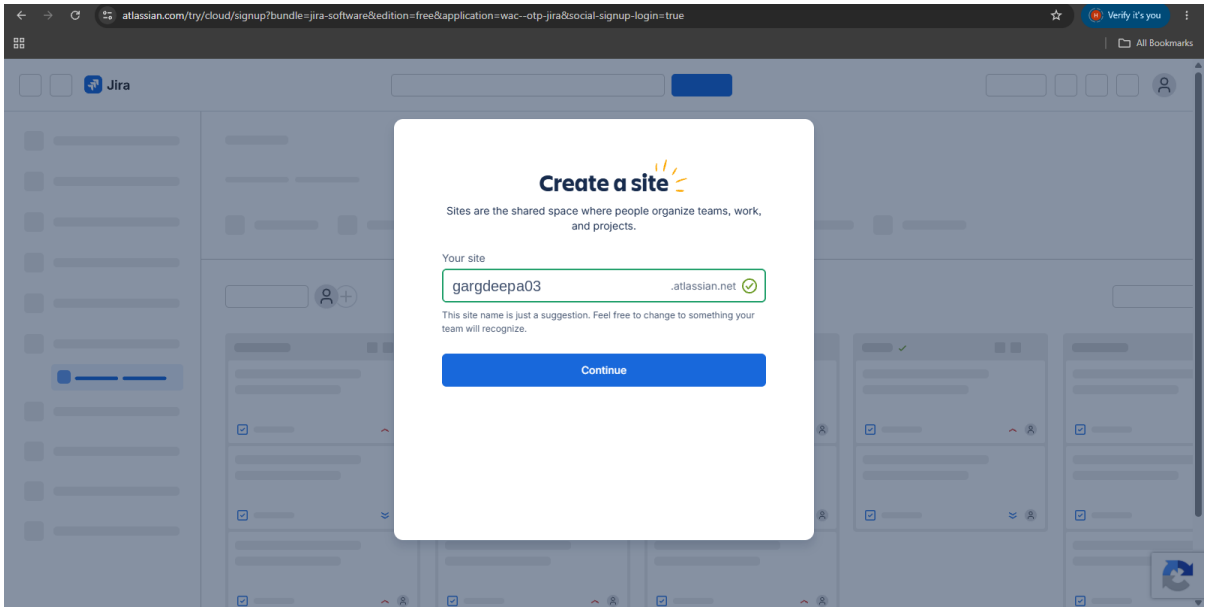
• Role-Based Access Control (RBAC): Access is governed at the organization and project levels via roles (Administrator, Member, Viewer).

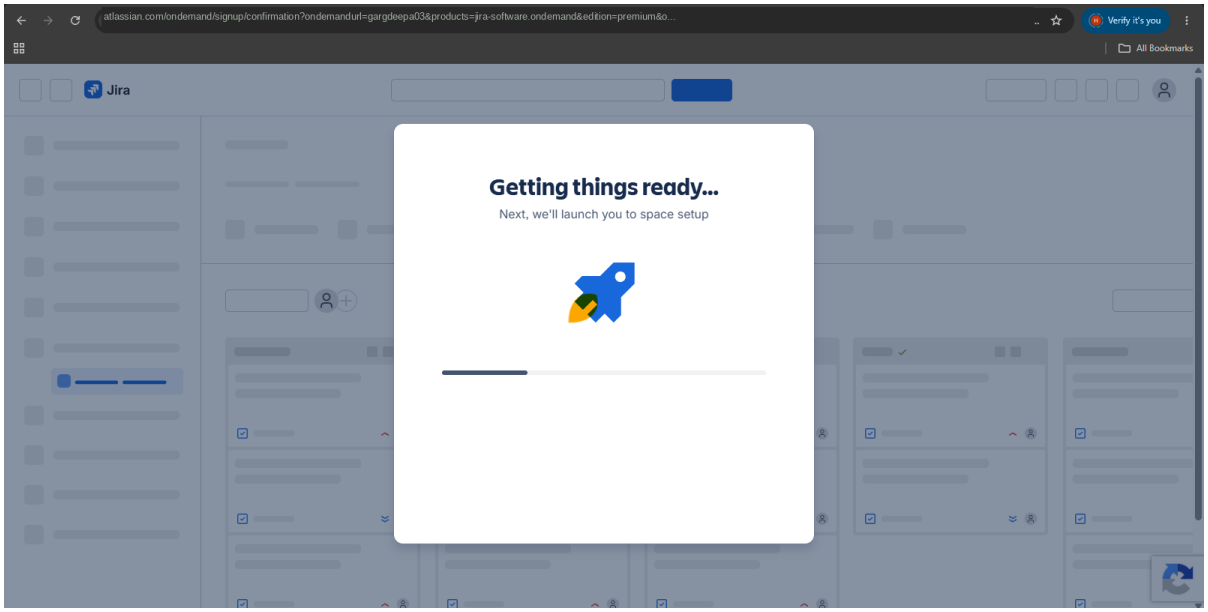
• Work Allocation: Tickets are assigned directly to individual developers, allowing visibility into sprint capacity, bottlenecks, and velocity reports.

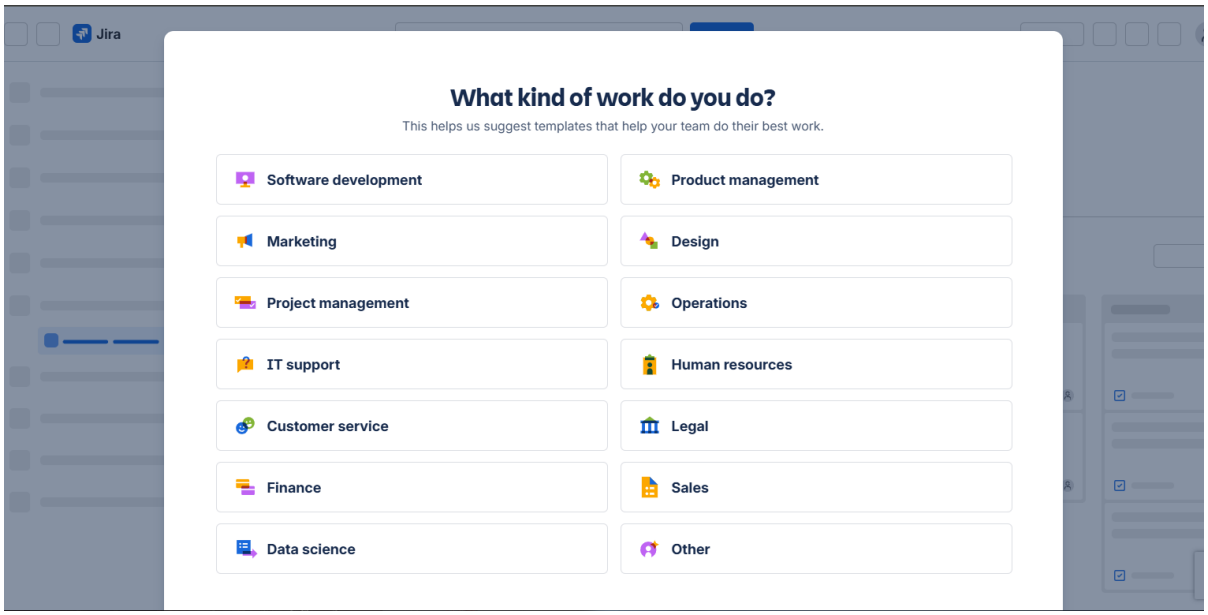
Procedure:

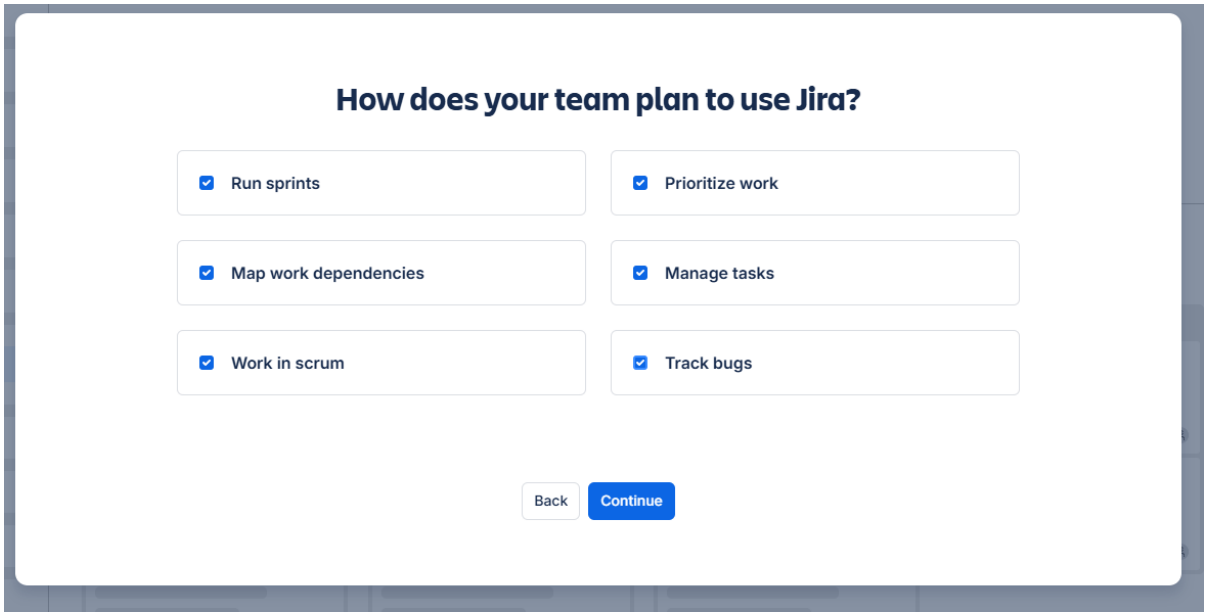
Step 1: Jira Cloud Account Signup & Initial Space Setup

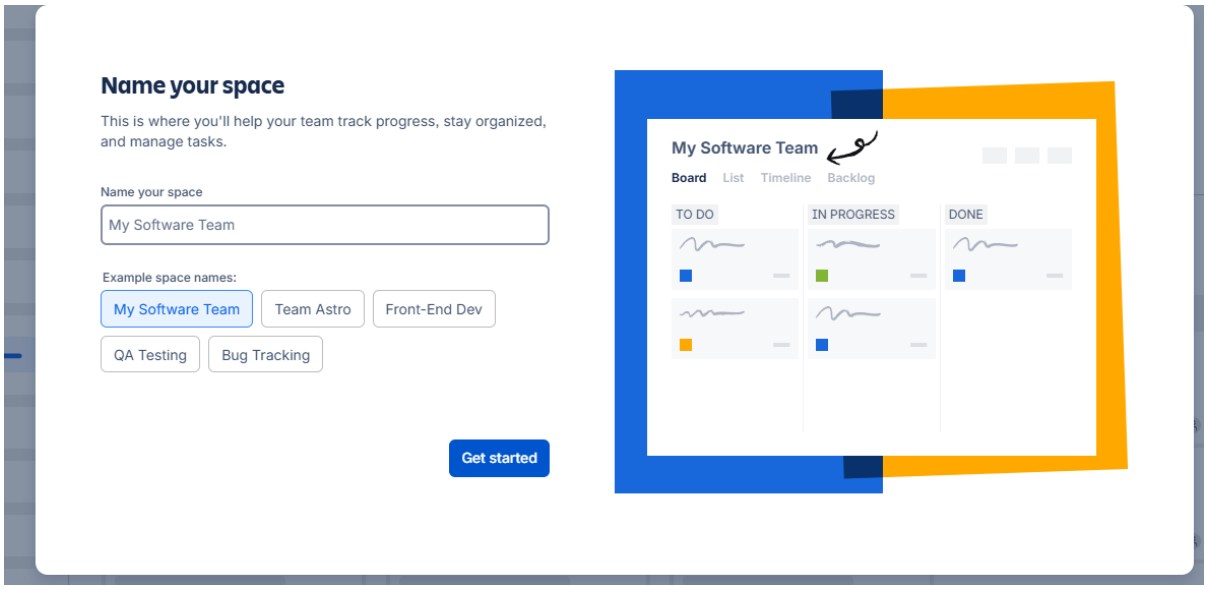
A new Jira Cloud site was created, a Software Development template was selected, and the initial team space ("My Software Team") was configured with a standard To Do / In Progress / In Review / Done workflow.

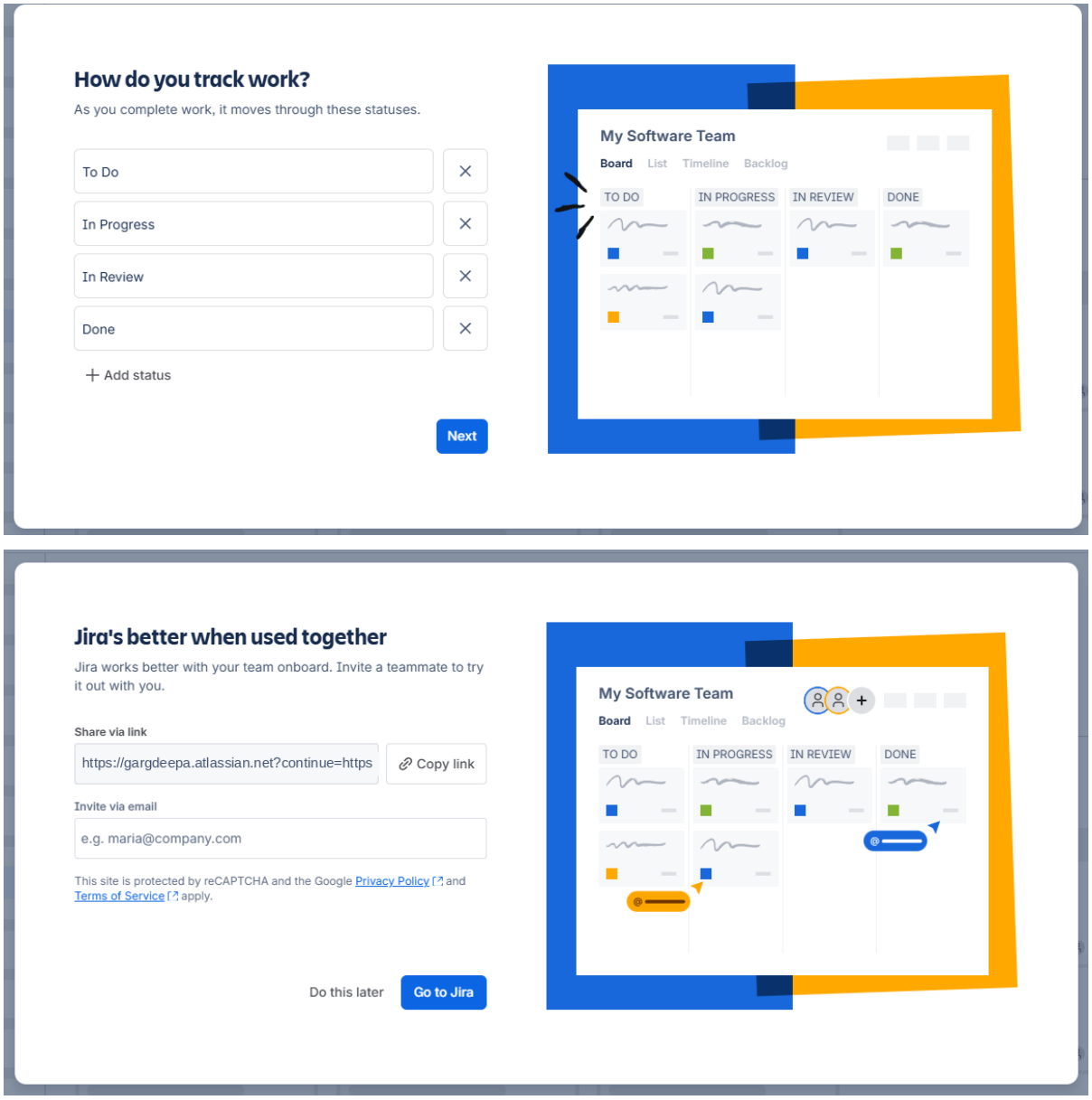



atlassian.com/ondemand/signup/confirmation?ondemandurl=gargdeepa03&products=jira-software.ondemand&edition=premium&o...
Verify it's you
All Bookmarks
Jira
Getting things ready...
Next, we'll launch you to space setup


Jira
What kind of work do you do?
This helps us suggest templates that help your team do their best work.
Software development
Product management
Marketing
Design
Project management
Operations
IT support
Human resources
Customer service
Legal
Finance
Sales
Data science
Other

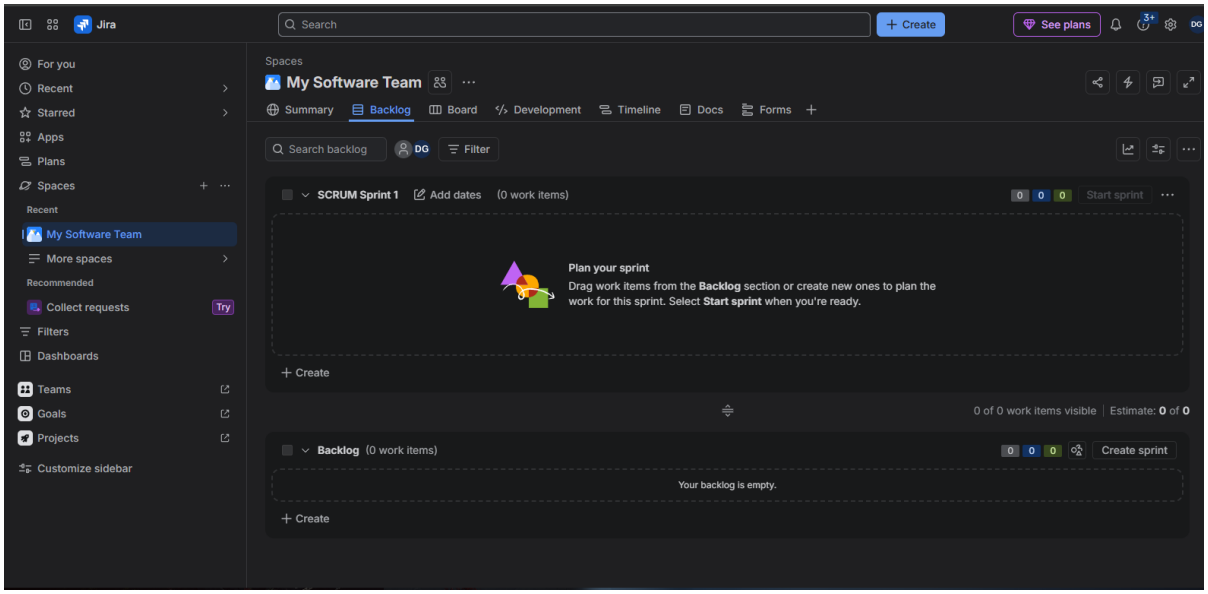

How does your team plan to use Jira?
Run sprints
Prioritize work
Map work dependencies
Manage tasks
Work in scrum
Track bugs
Back
Continue

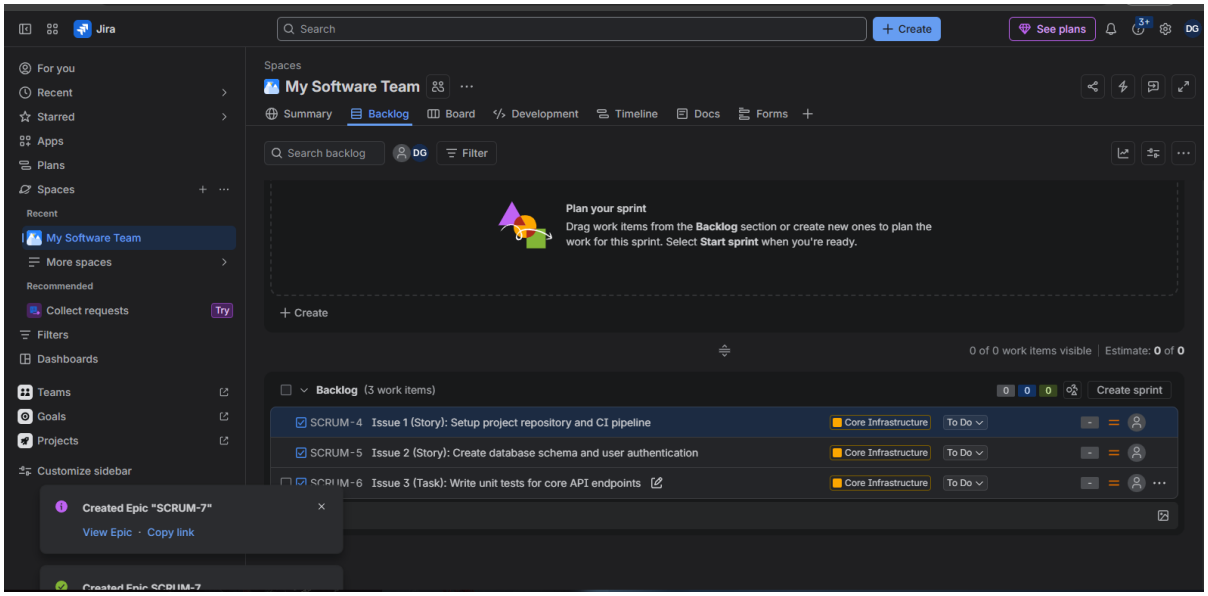

Name your space
This is where you'll help your team track progress, stay organized, and manage tasks.
Name your space
My Software Team
Example space names:
My Software Team
Team Astro
Front-End Dev
QA Testing
Bug Tracking
Get started
My Software Team
Board
List
Timeline
Backlog
TO DO
IN PROGRESS
DONE

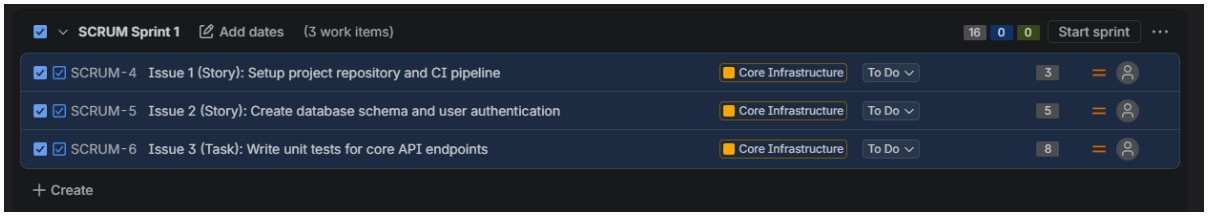


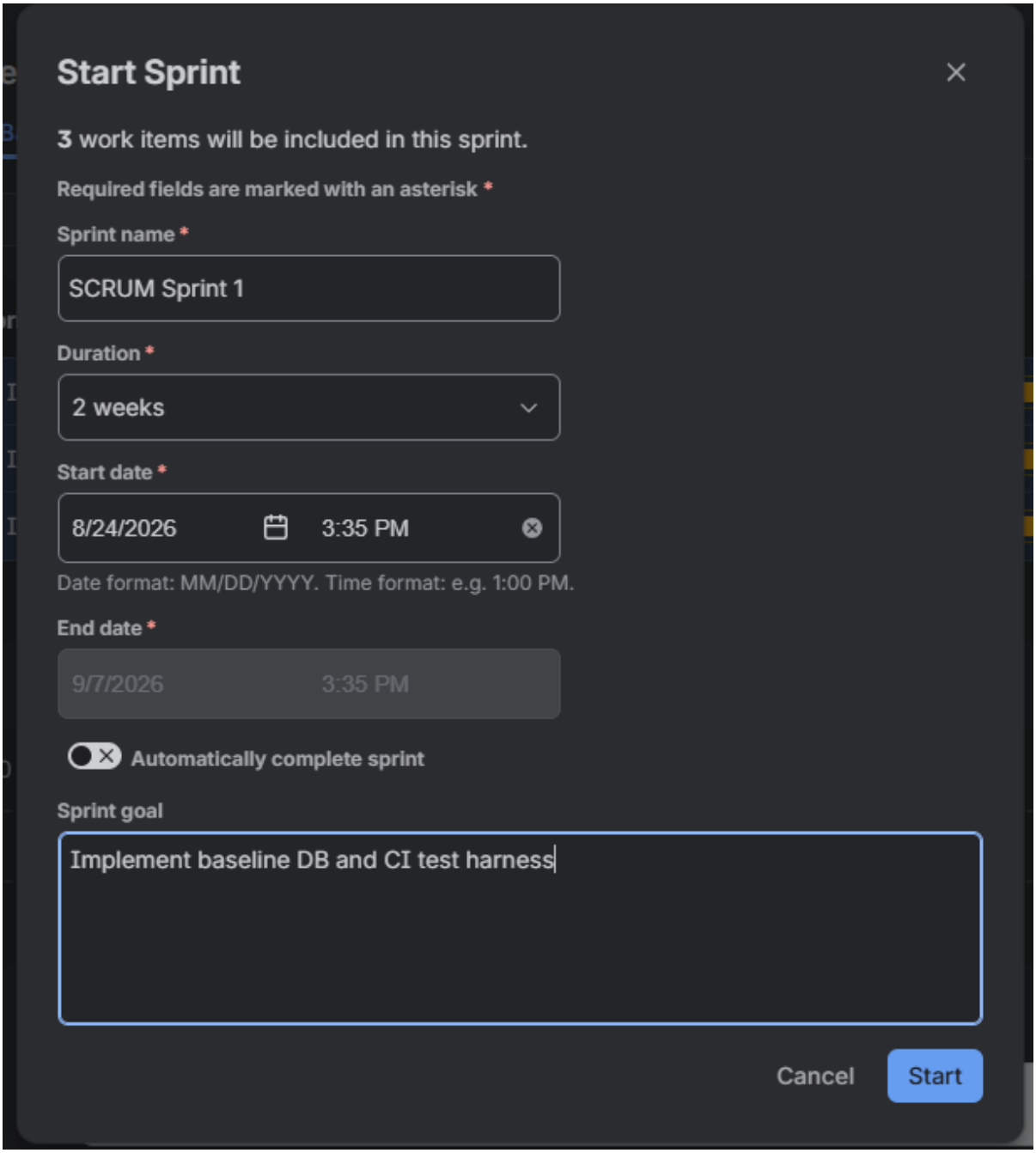
Step 2: Product Backlog Creation & Sprint Planning

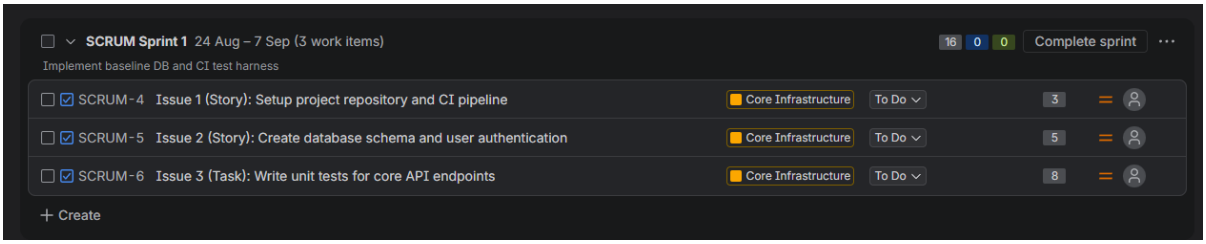
A SCRUM Sprint 1 container was created and three backlog items were added — two Stories and one Task, all mapped under the "Core Infrastructure" Epic.

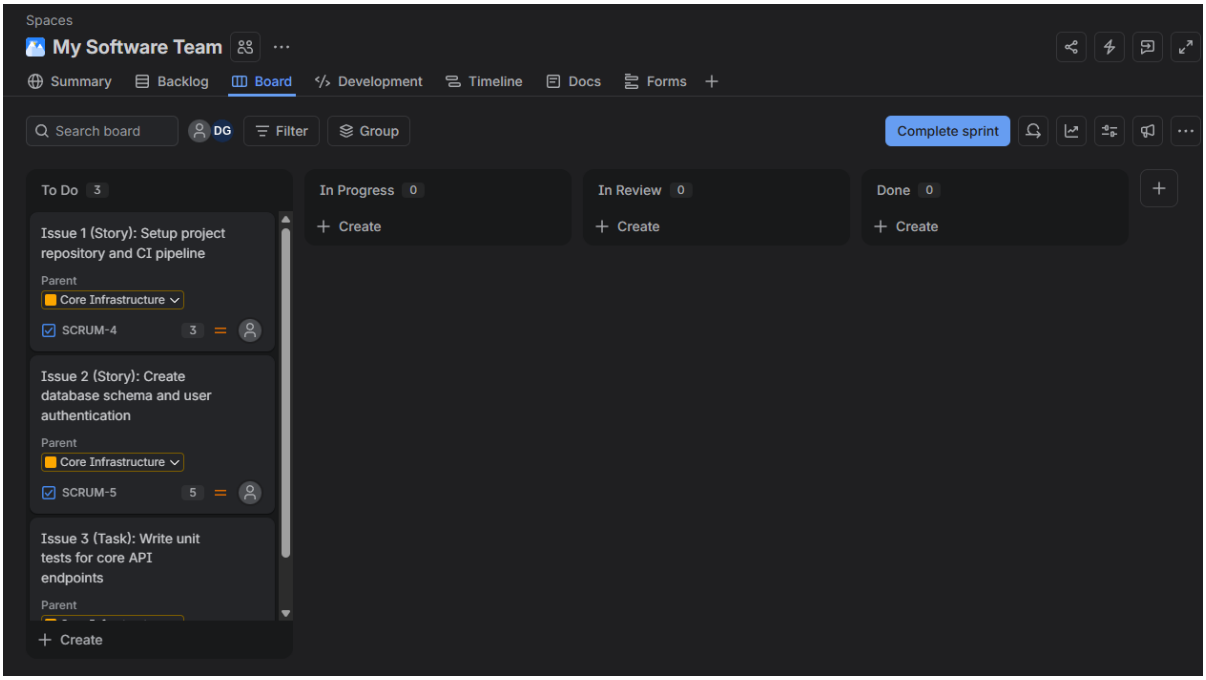

Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
My Software Team
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Search backlog
DG
Filter
SCRUM Sprint 1
Add dates
(0 work items)
Start sprint
Plan your sprint
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
Create
0 of 0 work items visible
Estimate: 0 of 0
Backlog (0 work items)
Create sprint
Your backlog is empty.
Create


Jira
Search
Create
See plans
Spaces
My Software Team
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Search backlog
DG
Filter
Plan your sprint
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
Create
0 of 0 work items visible
Estimate: 0 of 0
Backlog (3 work items)
Create sprint
SCRUM-4 Issue 1 (Story): Setup project repository and CI pipeline
SCRUM-5 Issue 2 (Story): Create database schema and user authentication
SCRUM-6 Issue 3 (Task): Write unit tests for core API endpoints
Core Infrastructure
To Do
Created Epic "SCRUM-7"
View Epic · Copy link


SCRUM Sprint 1
Add dates
(3 work items)
16
0
0
Start sprint
SCRUM-4 Issue 1 (Story): Setup project repository and CI pipeline
SCRUM-5 Issue 2 (Story): Create database schema and user authentication
SCRUM-6 Issue 3 (Task): Write unit tests for core API endpoints
Core Infrastructure
To Do
3
5
8
Create

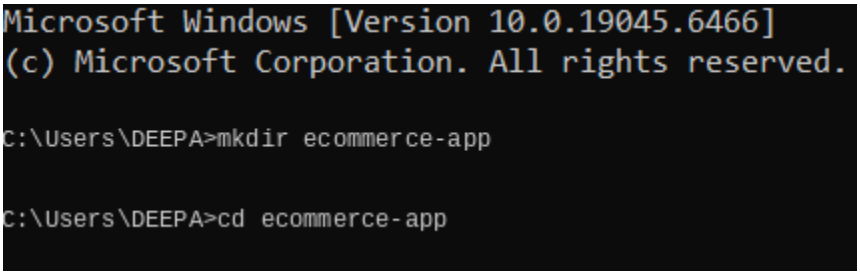

Start Sprint
3 work items will be included in this sprint.
Required fields are marked with an asterisk *
Sprint name *
SCRUM Sprint 1
Duration *
2 weeks
Start date *
8/24/2026
3:35 PM
Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.
End date *
9/7/2026
3:35 PM
Automatically complete sprint
Sprint goal
Implement baseline DB and CI test harness
Cancel
Start


SCRUM Sprint 1 24 Aug – 7 Sep (3 work items)
16 0 0
Complete sprint
Implement baseline DB and CI test harness
SCRUM-4 Issue 1 (Story): Setup project repository and CI pipeline
Core Infrastructure
To Do
3
SCRUM-5 Issue 2 (Story): Create database schema and user authentication
Core Infrastructure
To Do
5
SCRUM-6 Issue 3 (Task): Write unit tests for core API endpoints
Core Infrastructure
To Do
8
+ Create

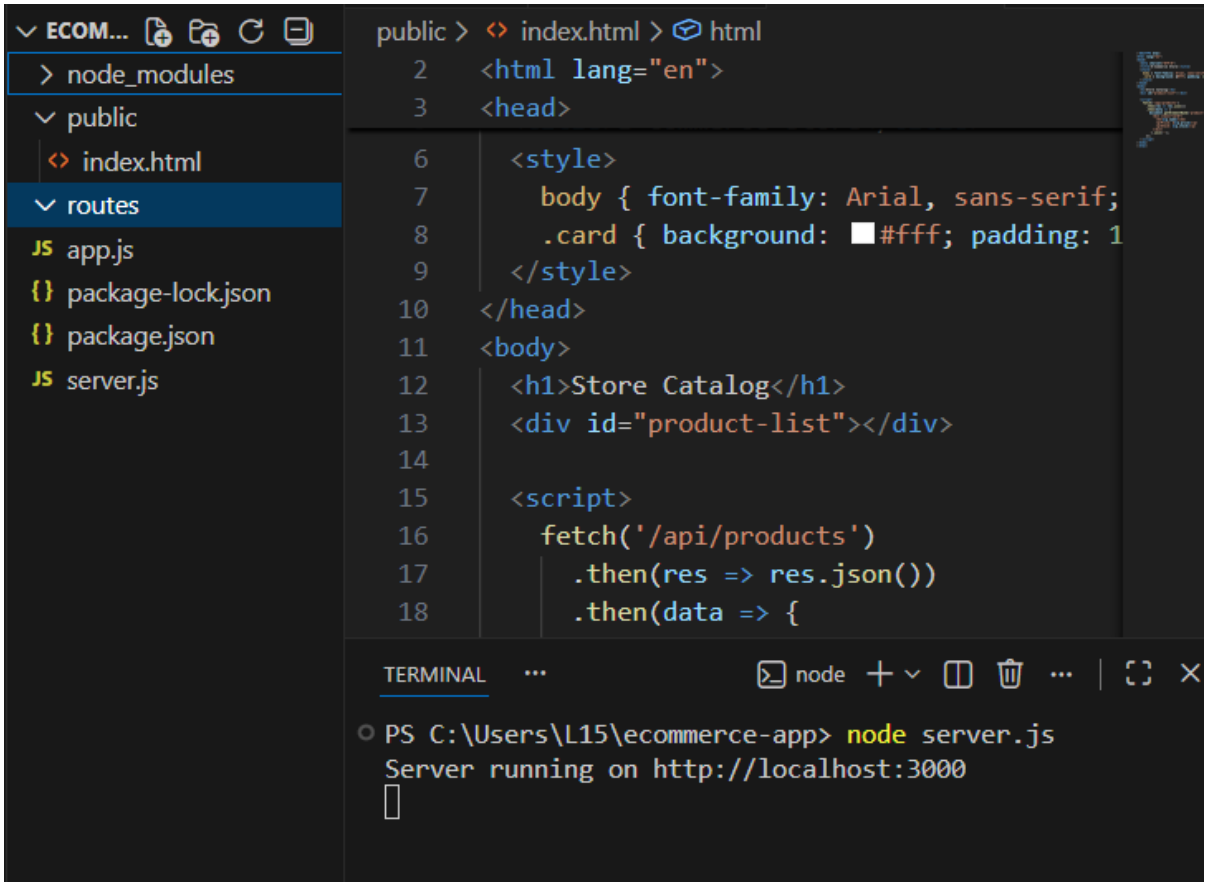


Step 3: Local Project Scaffolding (Node.js / Express)

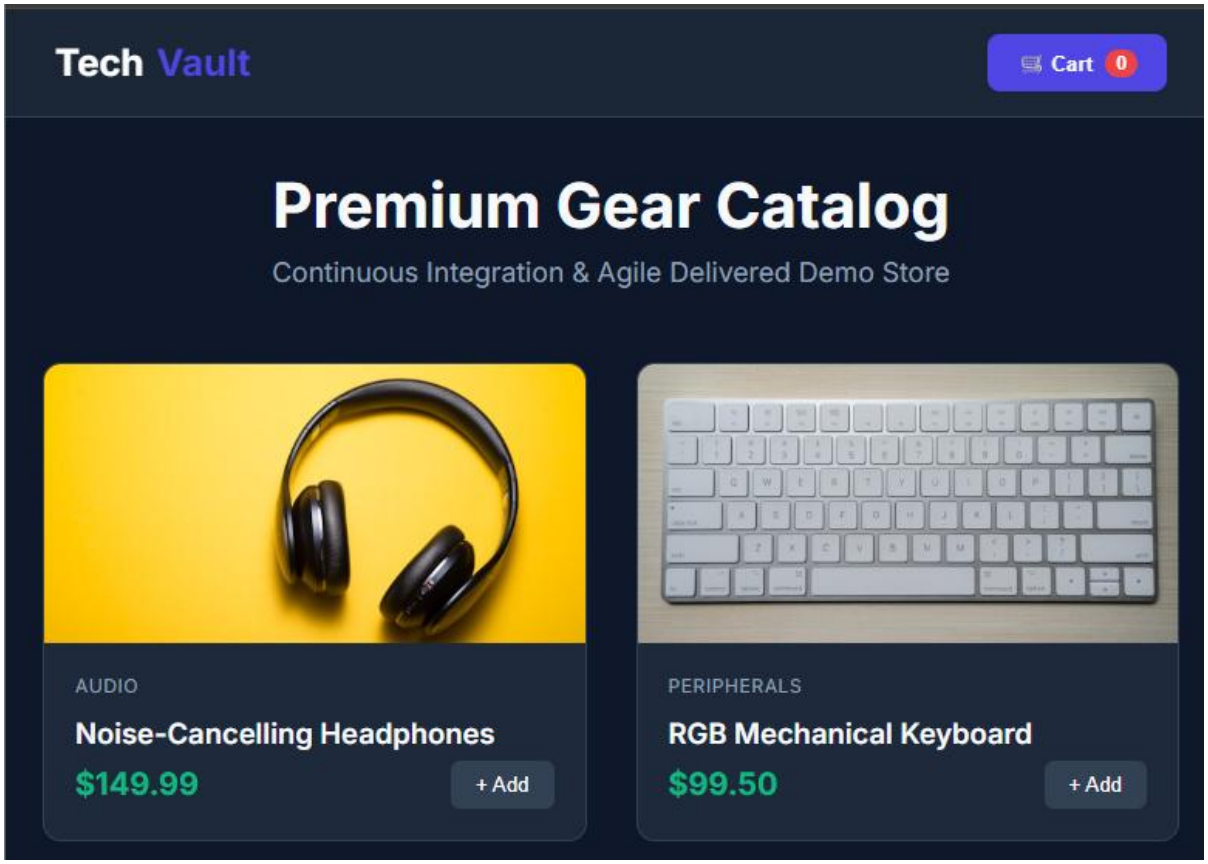
A local Node.js project (ecommerce-app) was initialized via npm, and a minimal Express-based storefront demo ("Tech Vault") was built to serve as the application under version control.

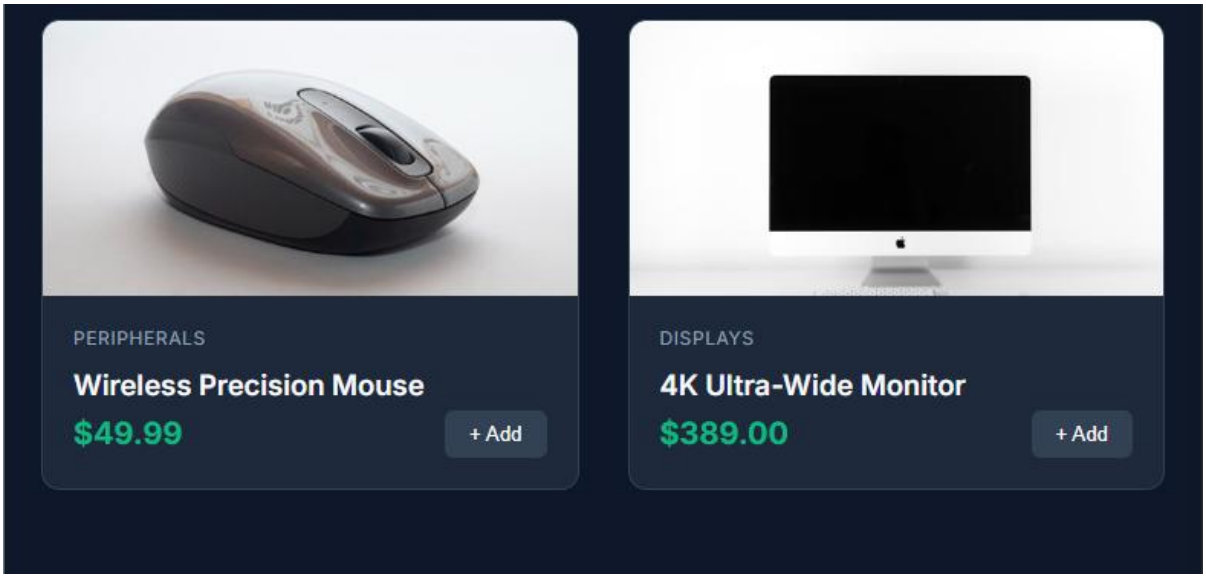





ECOM...
node_modules
public
index.html
routes
app.js
package-lock.json
package.json
server.js
public > index.html > html
<html lang="en">
<head>
<style>
body { font-family: Arial, sans-serif;
.card { background: #fff; padding: 1
</style>
</head>
<body>
<h1>Store Catalog</h1>

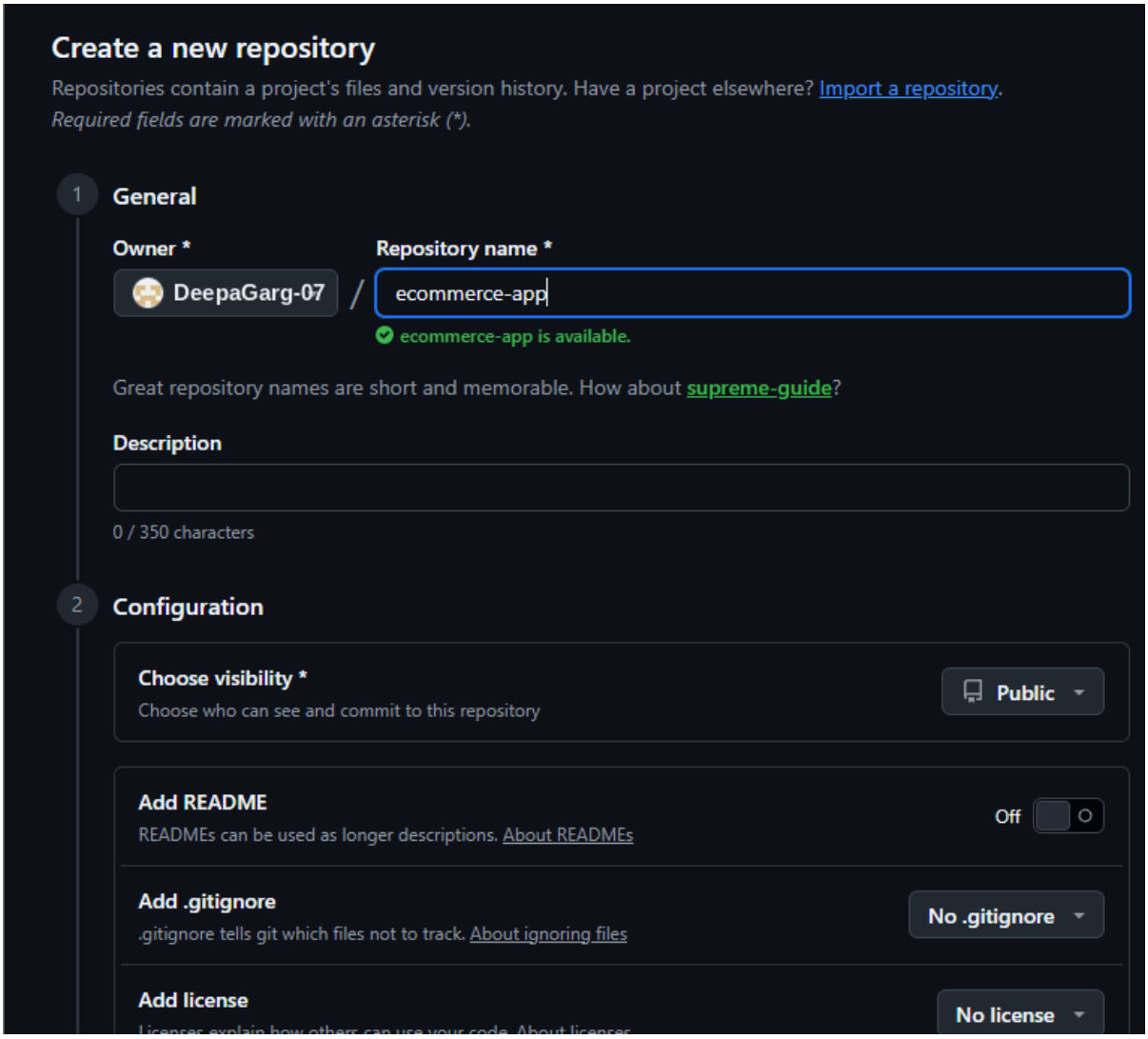
<script>
fetch('/api/products')
.then(res => res.json())
.then(data => {
TERMINAL
node
PS C:\Users\L15\ecommerce-app> node server.js
Server running on http://localhost:3000

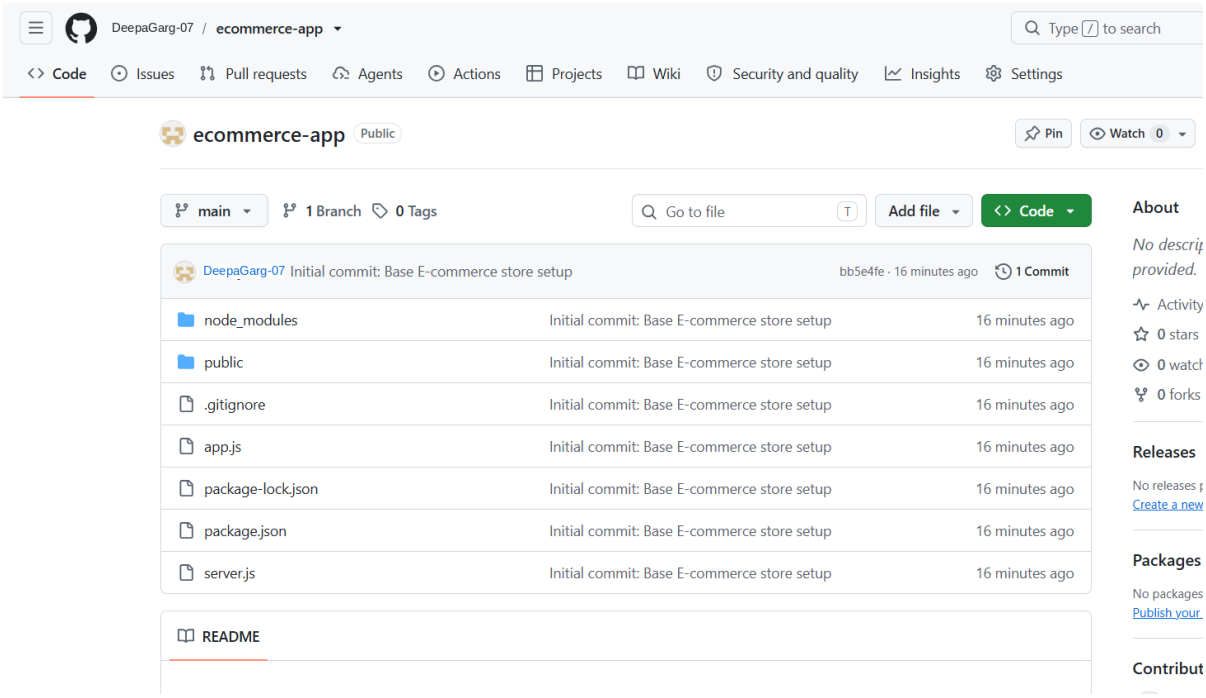

Tech Vault
Cart 0
Premium Gear Catalog
Continuous Integration & Agile Delivered Demo Store
AUDIO
Noise-Cancelling Headphones
$149.99
+ Add
PERIPHERALS
RGB Mechanical Keyboard
$99.50
+ Add

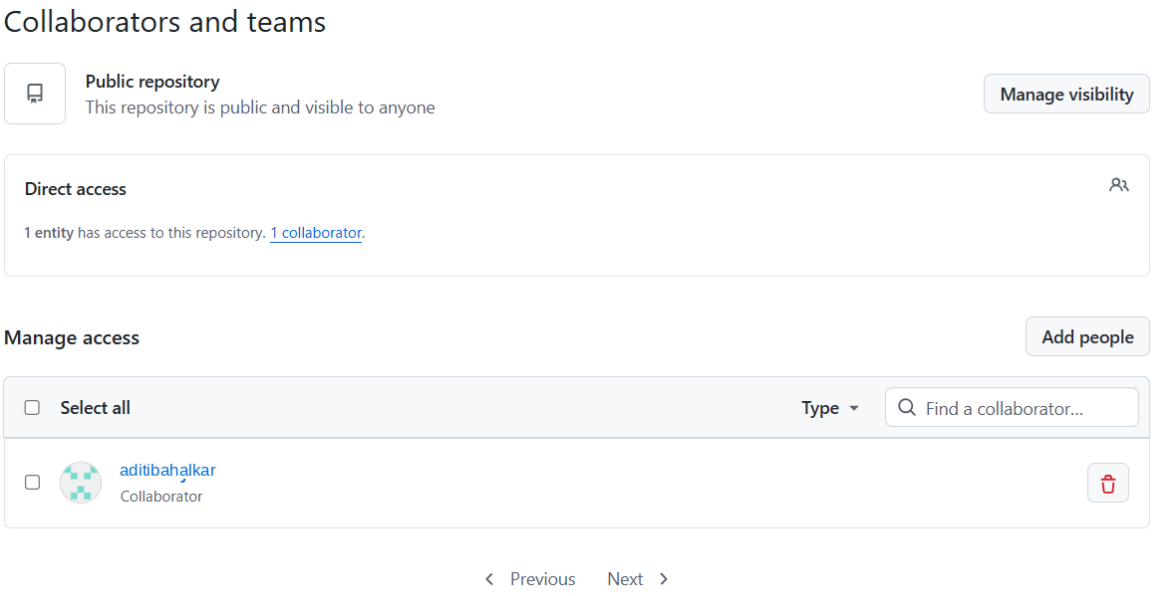


Step 4: GitHub Repository Creation & Collaborator Access

A public GitHub repository (DeepaGarg-07/ecommerce-app) was created for the project, and a collaborator was granted direct access to the repository.

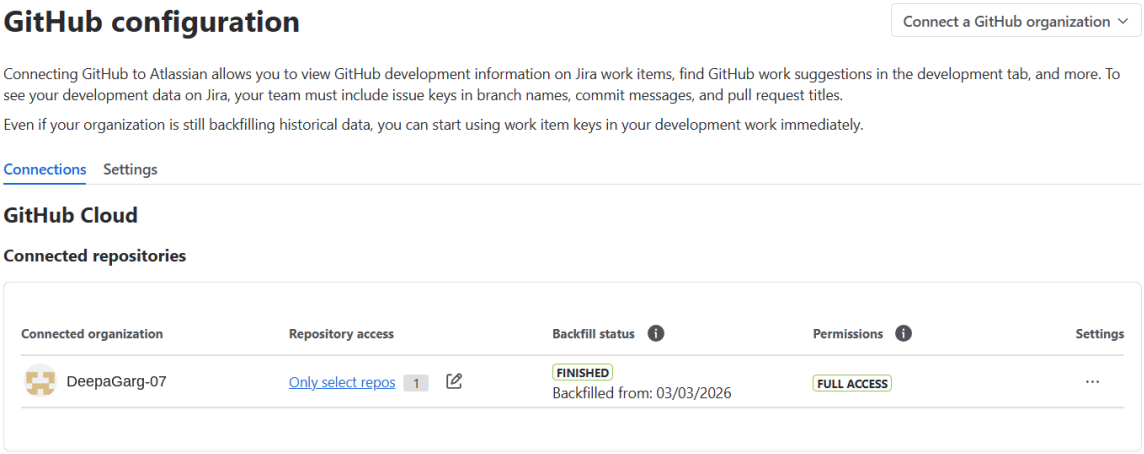

Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1 General
Owner *
Repository name *
DeepaGarg-07 / ecommerce-app
ecommerce-app is available.
Great repository names are short and memorable. How about supreme-guide?
Description
0 / 350 characters
2 Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
Off
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
No license


DeepaGarg-07 / ecommerce-app
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
ecommerce-app
Public
Pin
Watch 0
main
1 Branch
0 Tags
Go to file
Add file
Code
About
DeepaGarg-07 Initial commit: Base E-commerce store setup
bb5e4fe · 16 minutes ago
1 Commit
node_modules Initial commit: Base E-commerce store setup 16 minutes ago
public Initial commit: Base E-commerce store setup 16 minutes ago
.gitignore Initial commit: Base E-commerce store setup 16 minutes ago
app.js Initial commit: Base E-commerce store setup 16 minutes ago
package-lock.json Initial commit: Base E-commerce store setup 16 minutes ago
package.json Initial commit: Base E-commerce store setup 16 minutes ago
server.js Initial commit: Base E-commerce store setup 16 minutes ago
README
Releases
Packages
Contribut



Step 5: GitHub–Jira Integration

The GitHub for Jira app was connected to the Jira Cloud site, linking the DeepaGarg-07 organization and granting repository access so that development activity could be reflected on Jira issues.



Step 6: Smart Commit Workflow — Linking Commits to Jira Issues

A feature branch was created and a commit referencing the Jira issue key (SCRUM-5) was pushed to GitHub. The Jira issue's Development panel automatically reflected the linked commit.

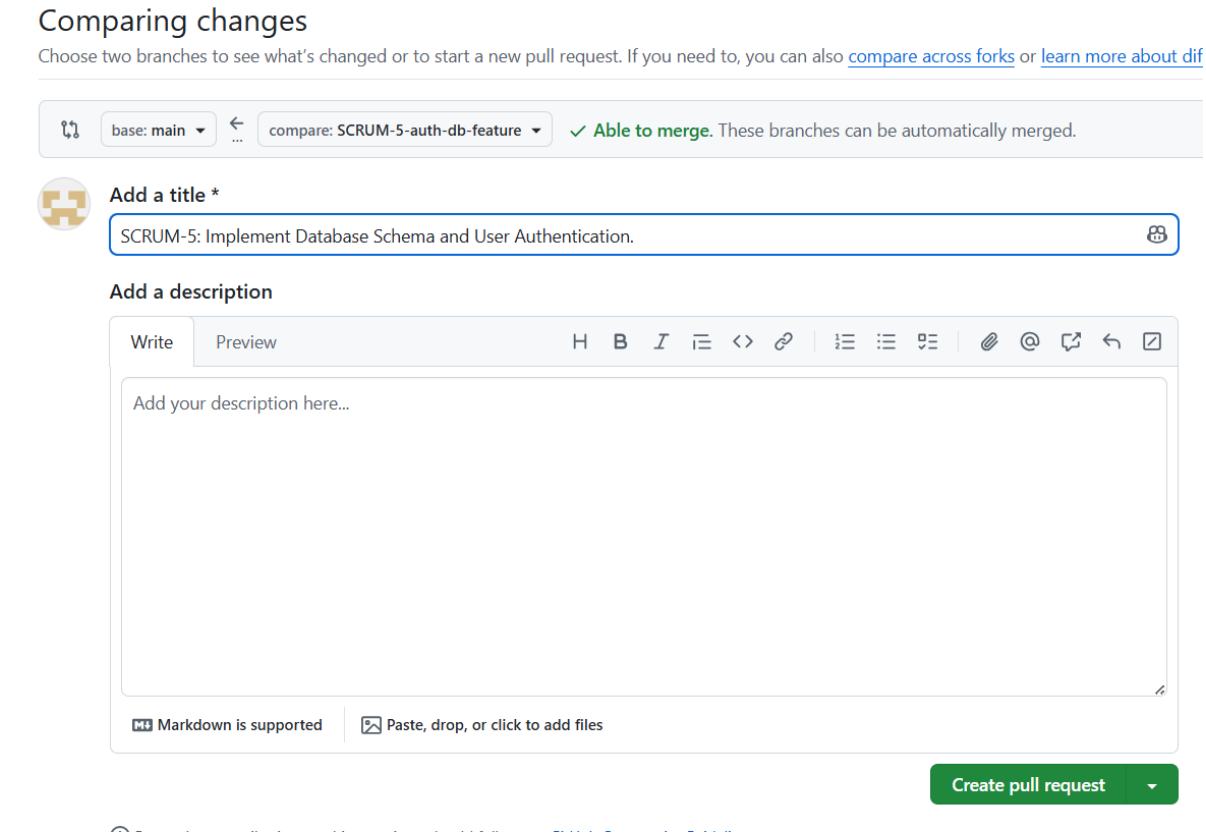
```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

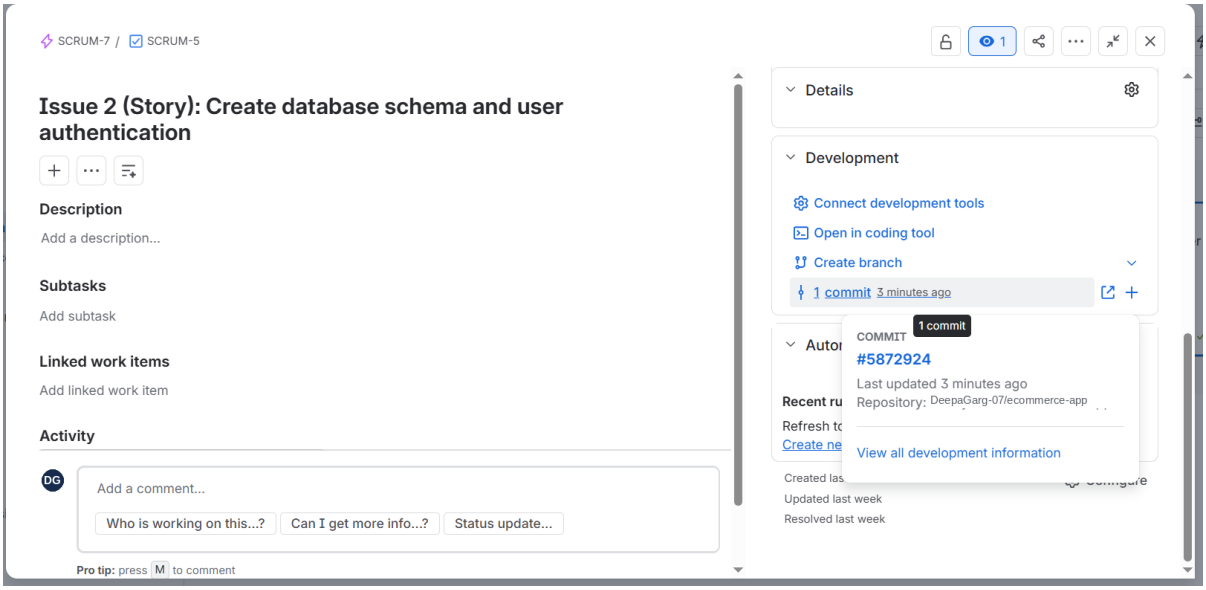
C:\Users\DEEPA\ecommerce-app>git checkout -b SCRUM-5-auth-db-feature
Switched to a new branch 'SCRUM-5-auth-db-feature'

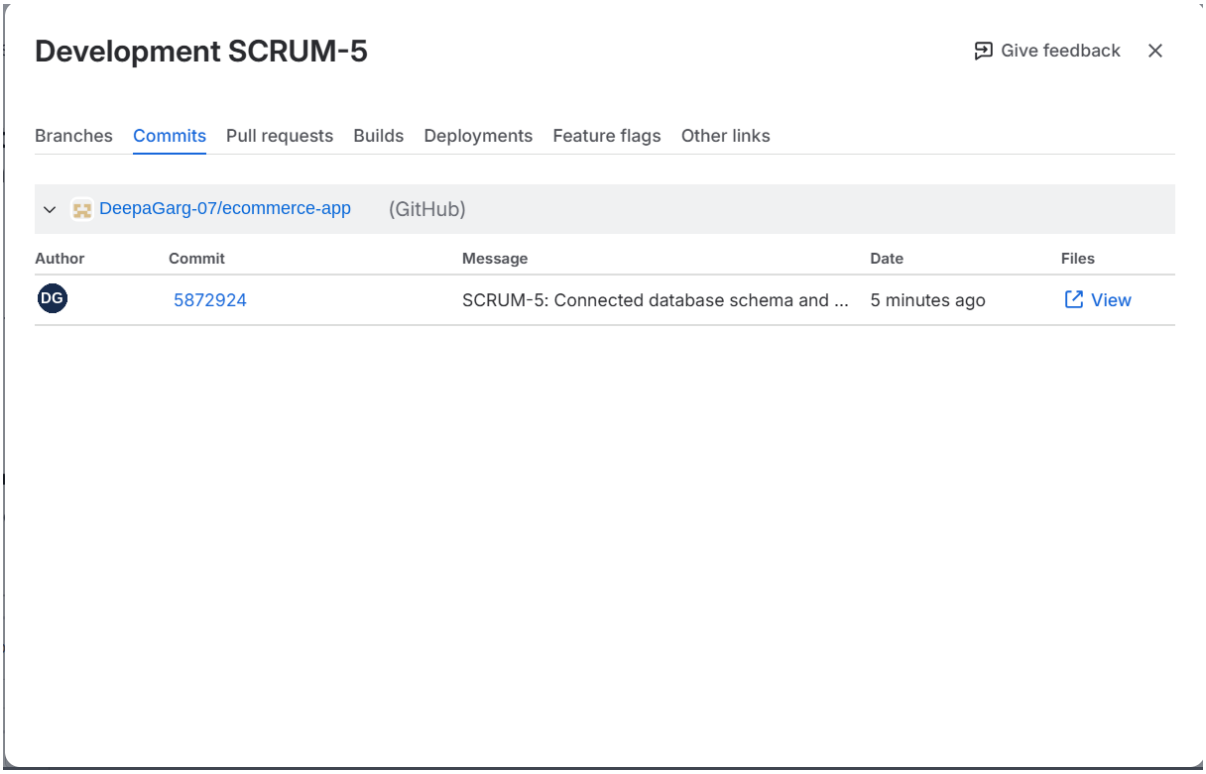
C:\Users\DEEPA\ecommerce-app>git commit --allow-empty -m "SCRUM-5: Connected database schema and user authentication structure"
[SCRUM-5-auth-db-feature 5872924] SCRUM-5: Connected database schema and user authentication structure

C:\Users\DEEPA\ecommerce-app>git push -u origin SCRUM-5-auth-db-feature
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 229 bytes | 229.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'SCRUM-5-auth-db-feature' on GitHub by visiting:
remote:       https://github.com/DeepaGarg-07/ecommerce-app/pull/new/SCRUM-5-auth-db-feature
remote:
To https://github.com/DeepaGarg-07/ecommerce-app.git
 * [new branch]      SCRUM-5-auth-db-feature -> SCRUM-5-auth-db-feature
branch 'SCRUM-5-auth-db-feature' set up to track 'origin/SCRUM-5-auth-db-feature'.

C:\Users\DEEPA\ecommerce-app>
```

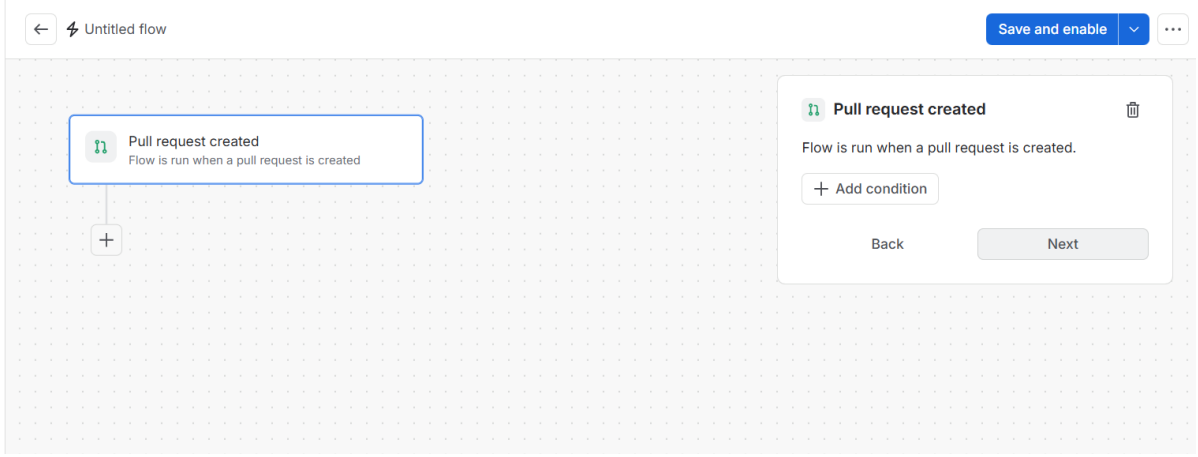


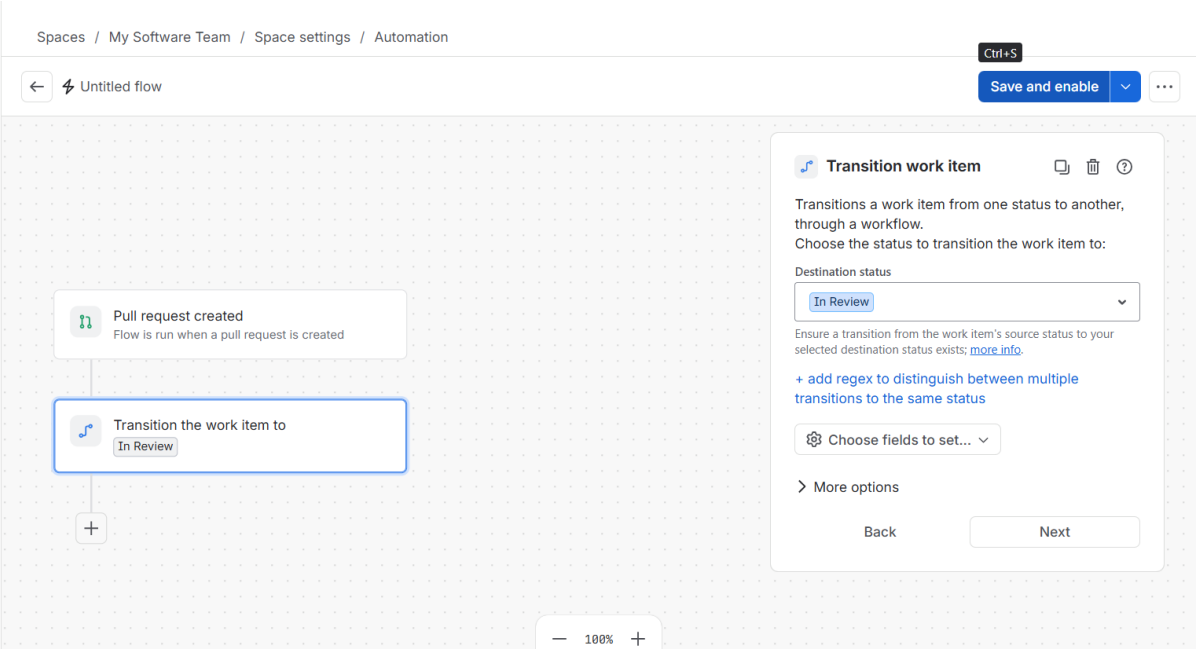


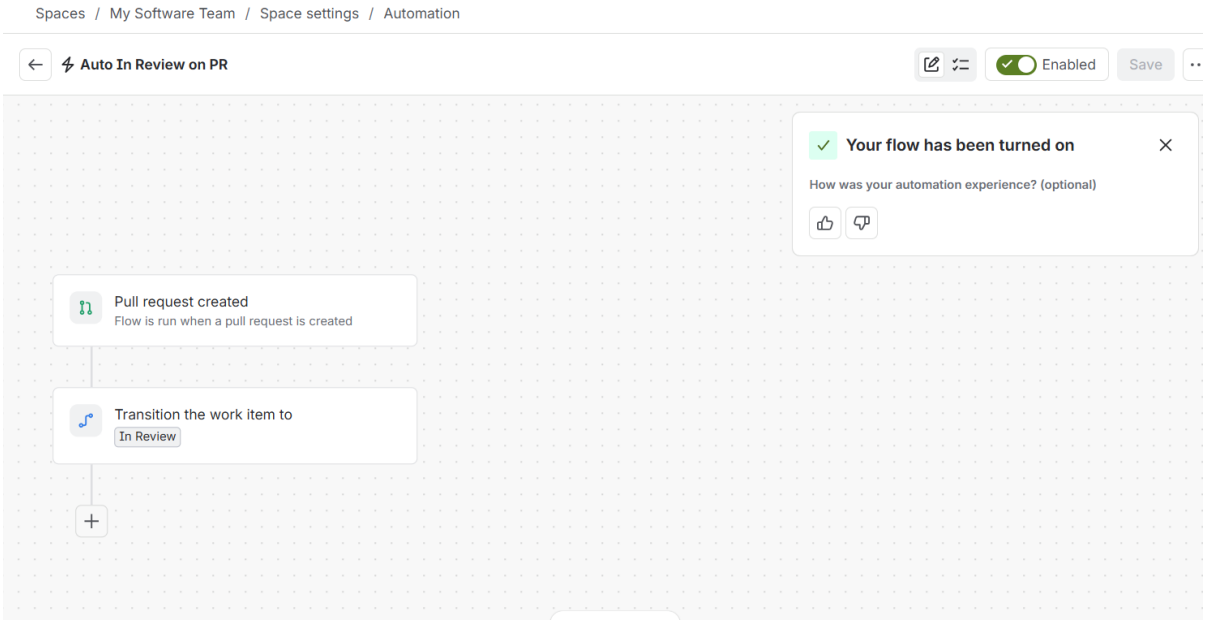


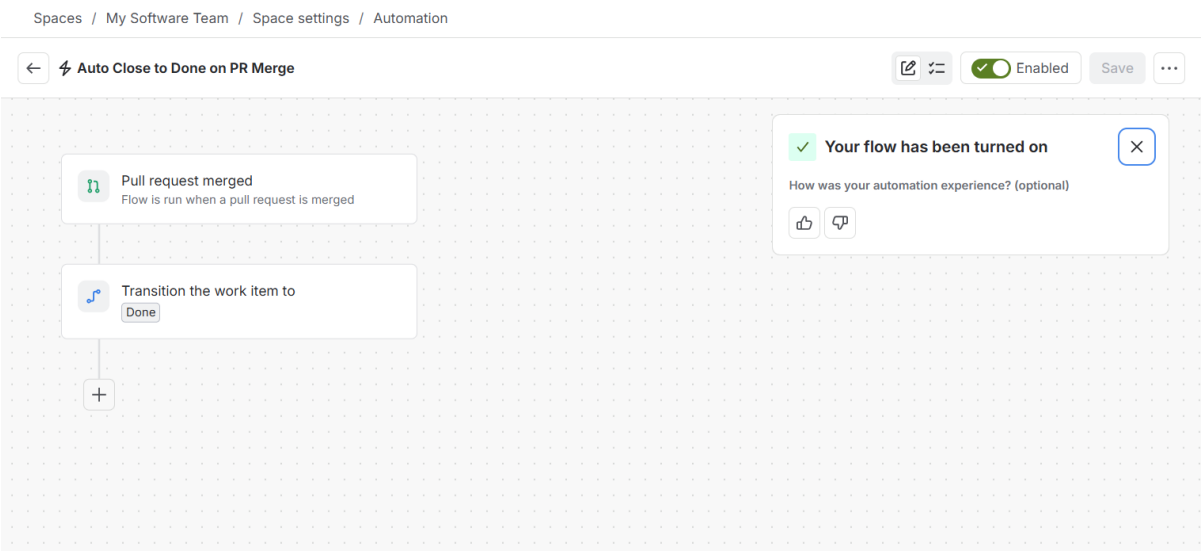
Step 7: Configuring Jira Automation Rules

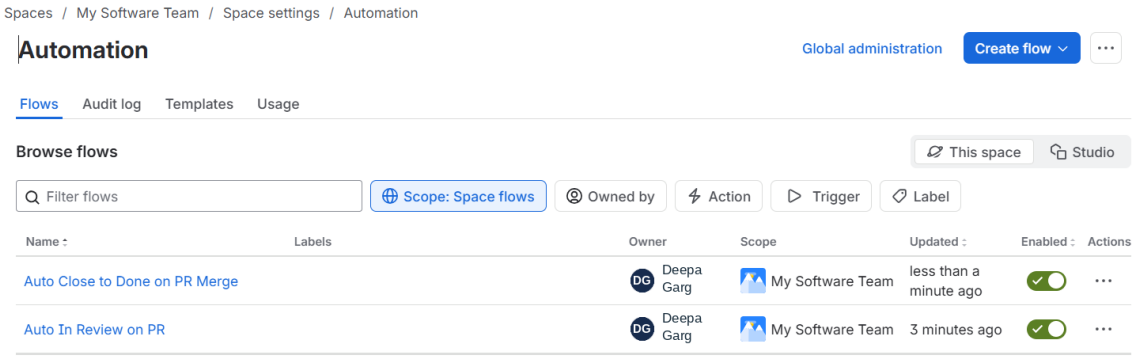
Two automation flows were built in Jira: one that transitions a work item to "In Review" when a Pull Request is created, and another that transitions it to "Done" when the Pull Request is merged.


Untitled flow
Save and enable
Pull request created
Flow is run when a pull request is created
Pull request created
Flow is run when a pull request is created.
Add condition
Back
Next


Spaces / My Software Team / Space settings / Automation
Untitled flow
Ctrl+S
Save and enable
Pull request created
Flow is run when a pull request is created
Transition the work item to
In Review
Transition work item
Transitions a work item from one status to another, through a workflow.
Choose the status to transition the work item to:
Destination status
In Review
Ensure a transition from the work item's source status to your selected destination status exists; more info.
+ add regex to distinguish between multiple transitions to the same status
Choose fields to set...
More options
Back
Next
100%

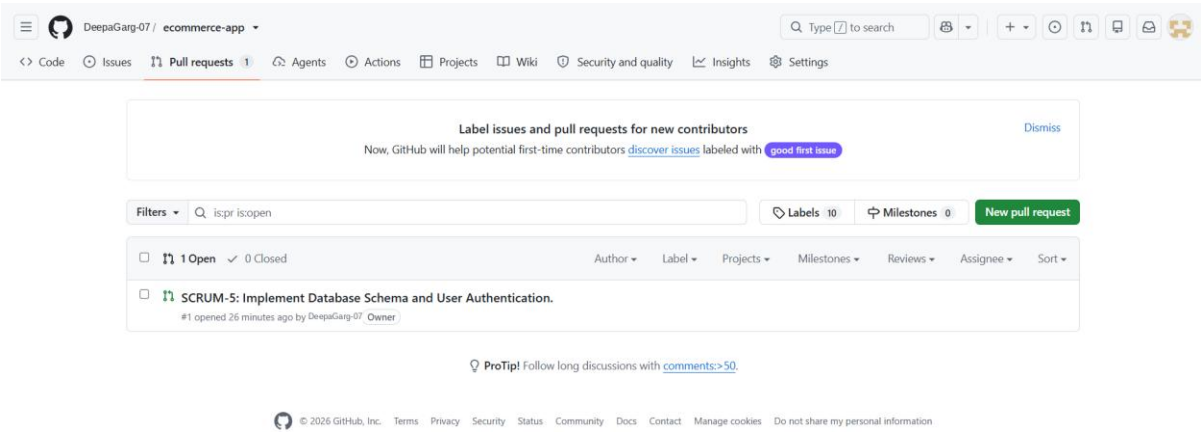

Spaces / My Software Team / Space settings / Automation
Auto In Review on PR
Enabled
Save
Your flow has been turned on
How was your automation experience? (optional)
Pull request created
Flow is run when a pull request is created
Transition the work item to
In Review

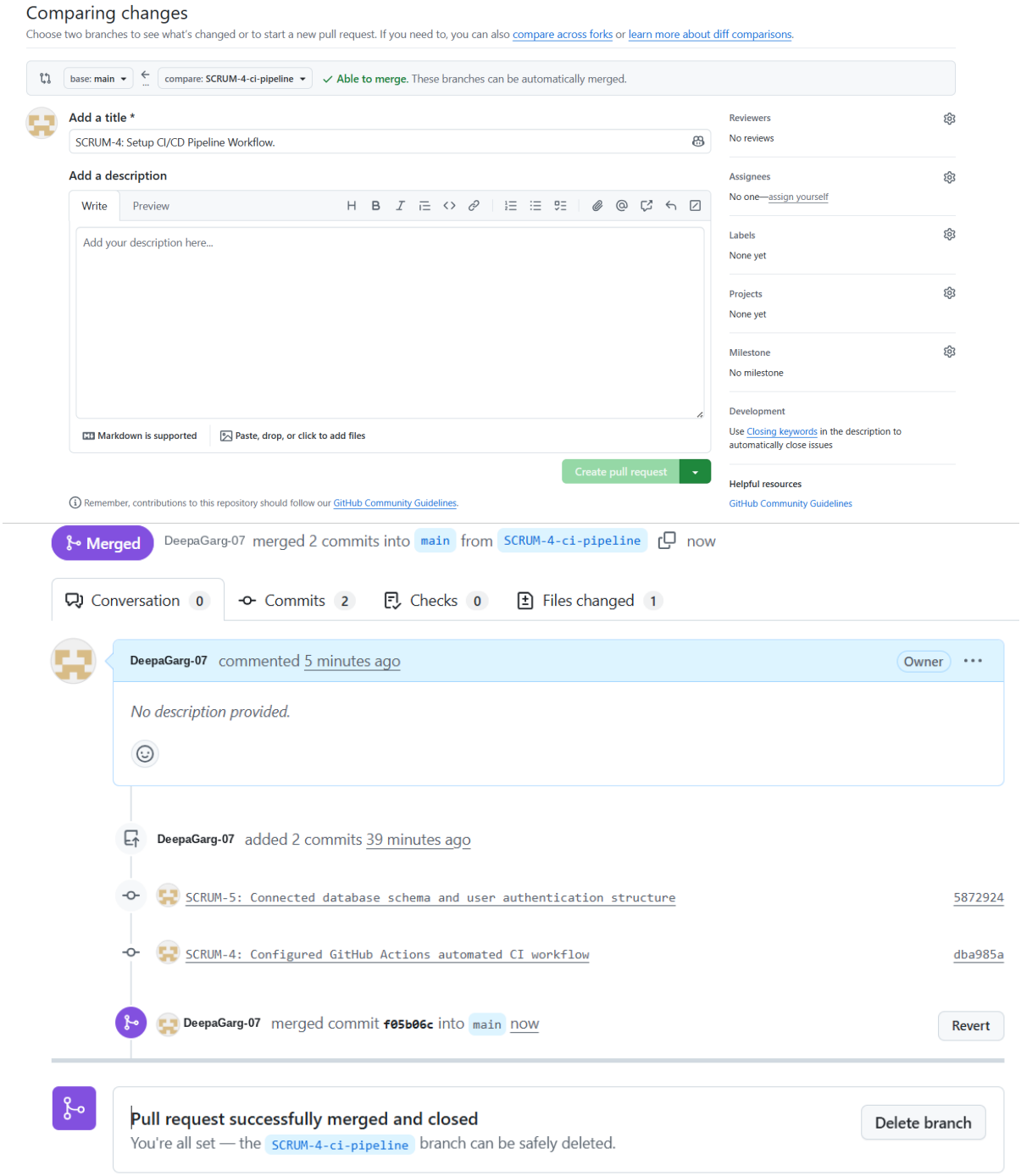


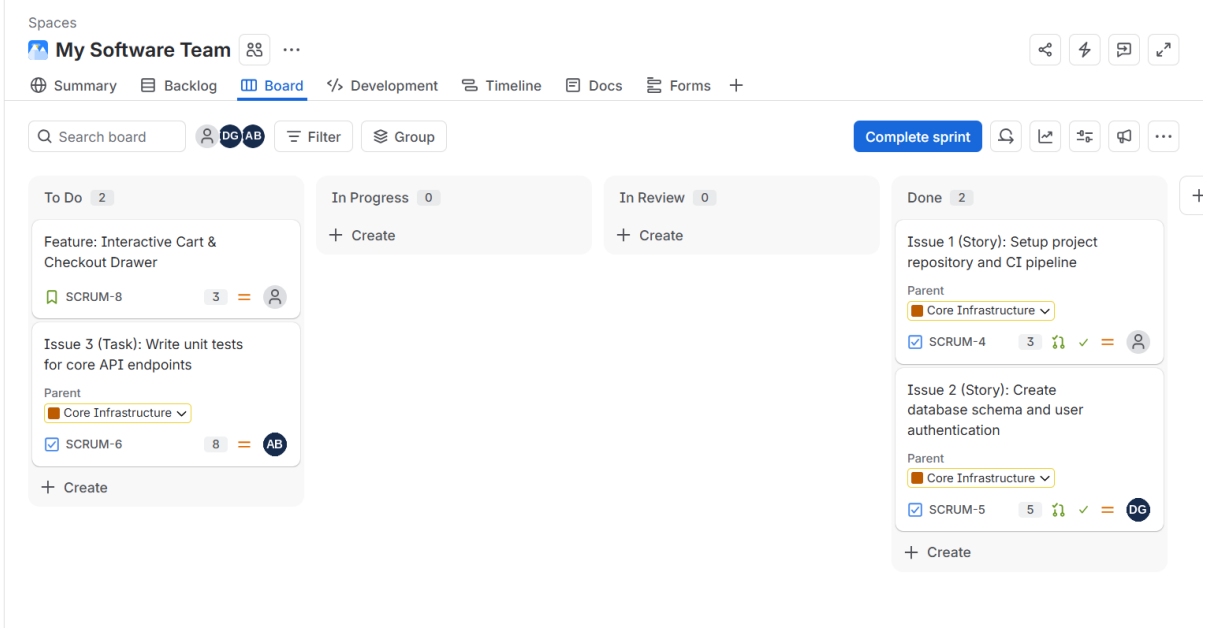


Step 8: Pull Request Workflow & Automated Status Transitions

A Pull Request (SCRUM-5) was opened, reviewed, and merged into main. The corresponding Jira issue automatically transitioned status via the configured automation rules, without any manual board updates.



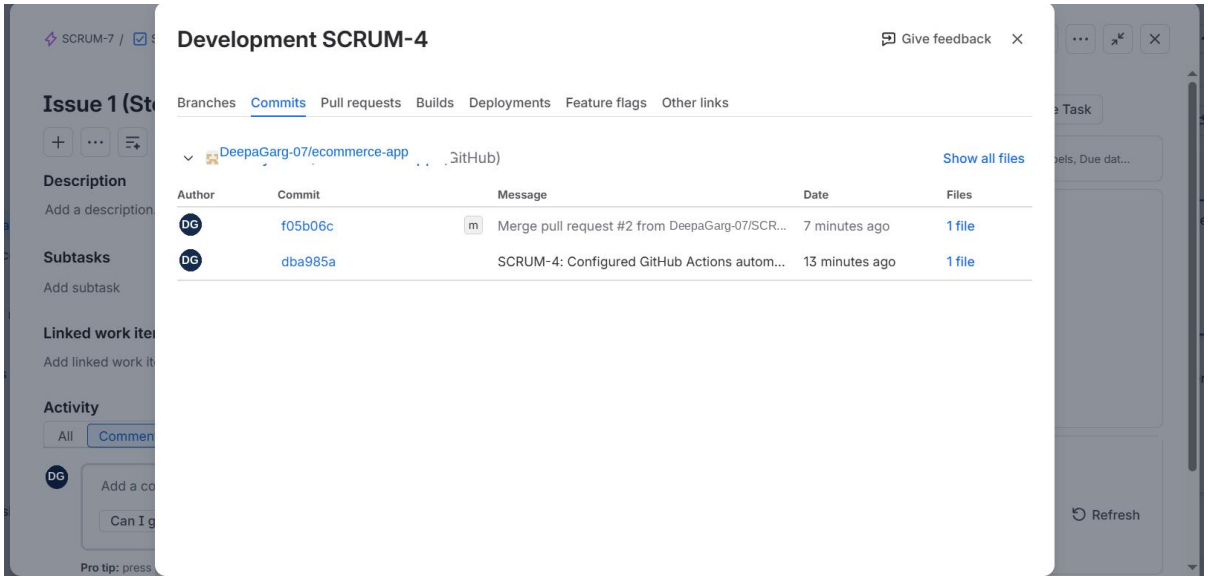

Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base: main
compare: SCRUM-4-ci-pipeline
Able to merge. These branches can be automatically merged.
Add a title *
SCRUM-4: Setup CI/CD Pipeline Workflow.
Add a description
Write
Preview
Add your description here...
Markdown is supported
Paste, drop, or click to add files
Create pull request
Remember, contributions to this repository should follow our GitHub Community Guidelines.
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Use Closing keywords in the description to automatically close issues
Helpful resources
GitHub Community Guidelines
Merged
DeepaGarg-07 merged 2 commits into main from SCRUM-4-ci-pipeline now
Conversation 0
Commits 2
Checks 0
Files changed 1
DeepaGarg-07 commented 5 minutes ago
Owner
No description provided.
DeepaGarg-07 added 2 commits 39 minutes ago
SCRUM-5: Connected database schema and user authentication structure 5872924
SCRUM-4: Configured GitHub Actions automated CI workflow dba985a
DeepaGarg-07 merged commit f05b06c into main now
Revert
Pull request successfully merged and closed
You're all set — the SCRUM-4-ci-pipeline branch can be safely deleted.
Delete branch

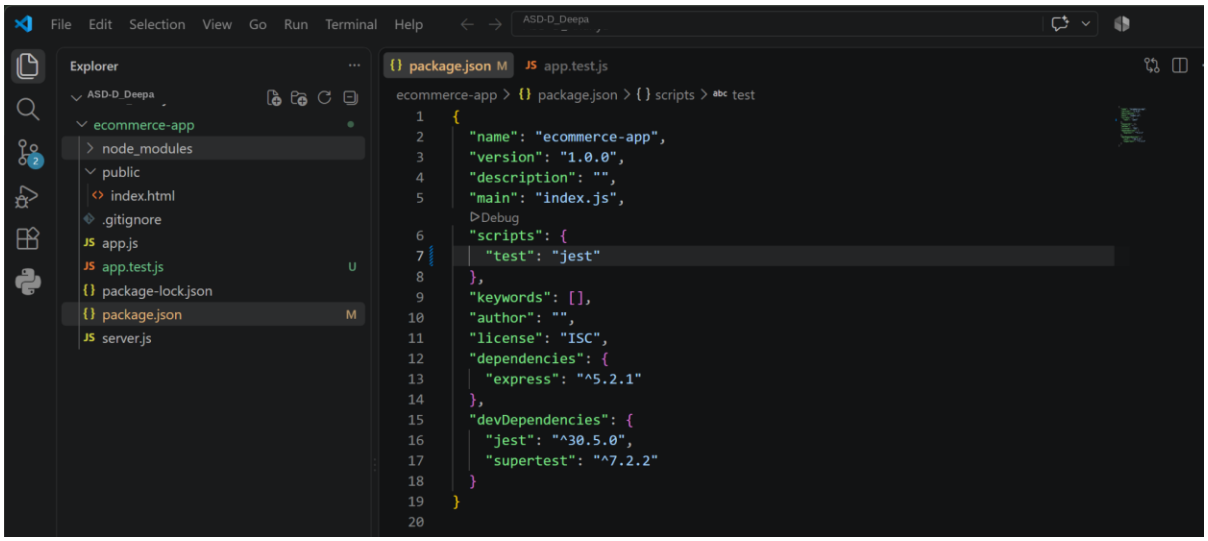


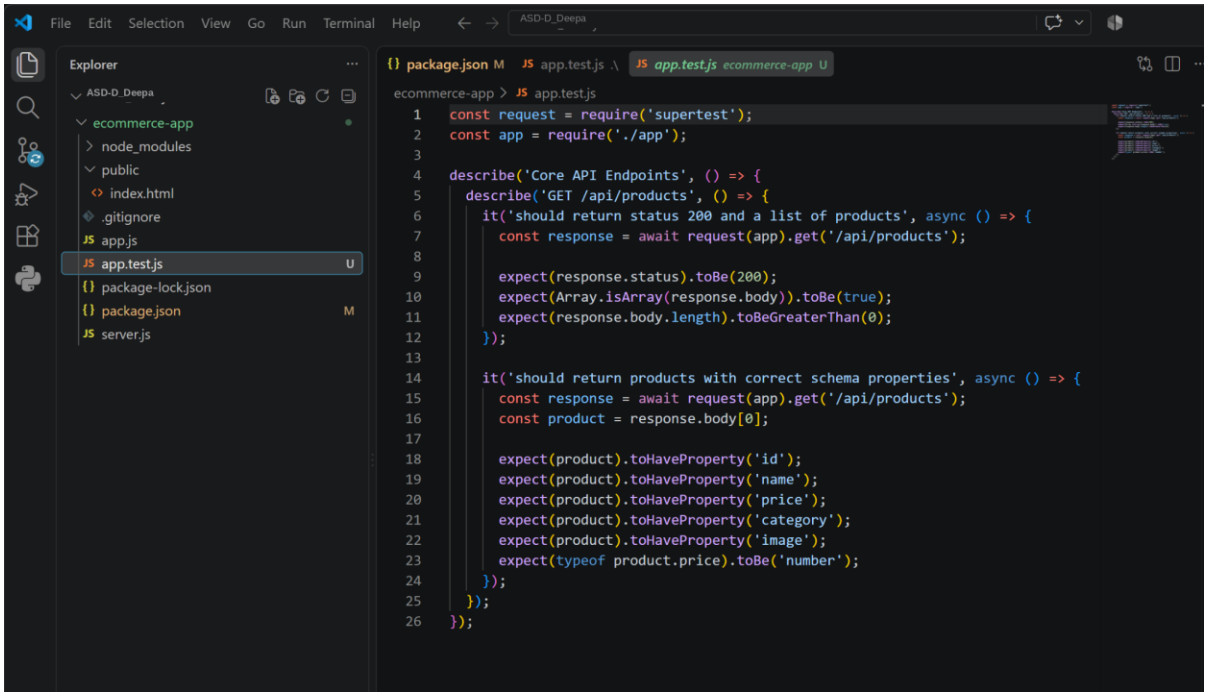


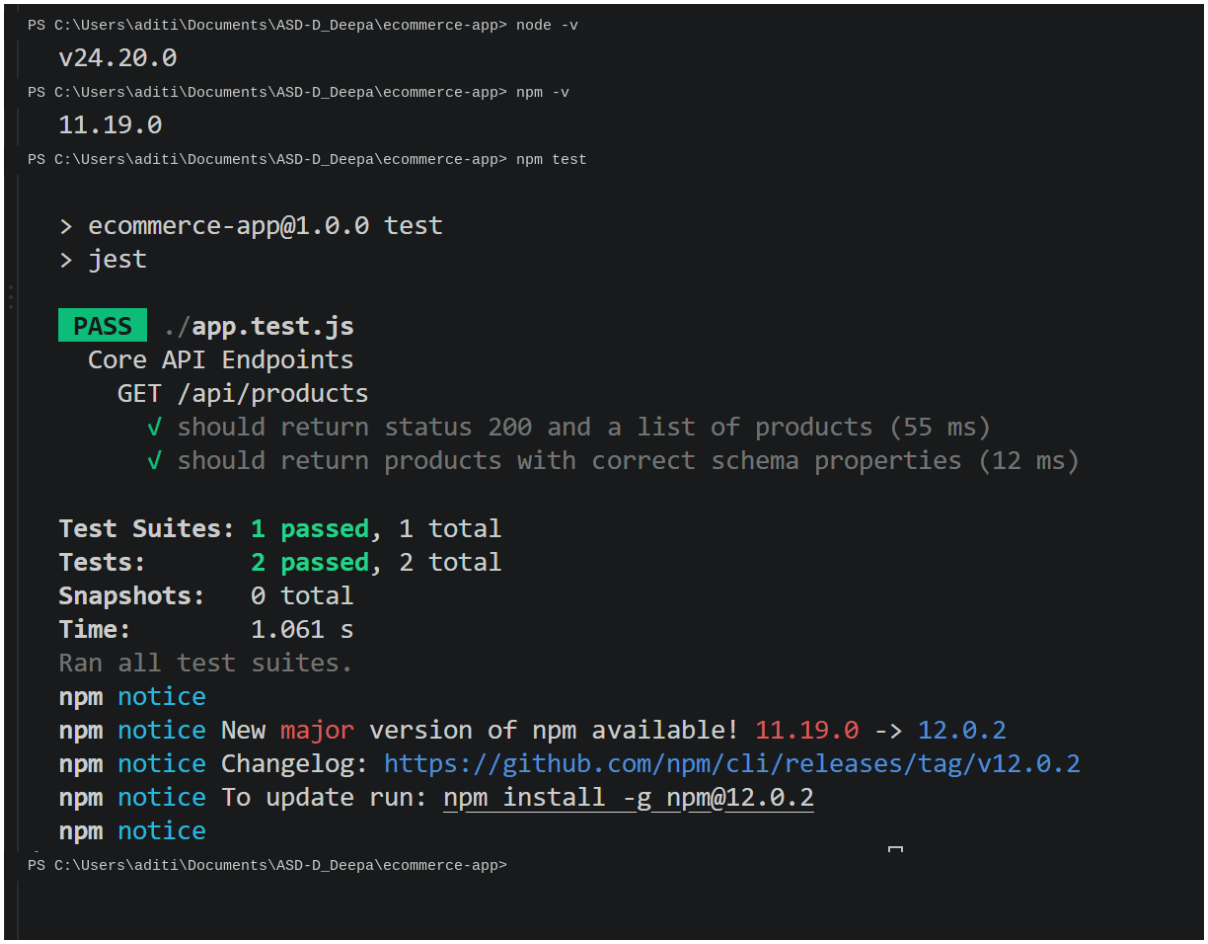
Step 9: CI/CD Pipeline Setup with GitHub Actions

A GitHub Actions workflow (SCRUM-4) was configured to automate the CI pipeline, and unit tests were written locally using Jest and Supertest to validate the core product API endpoints.


Development SCRUM-4
Give feedback
Branches
Commits
Pull requests
Builds
Deployments
Feature flags
Other links
DeepaGarg-07/ecommerce-app
Show all files
Author
Commit
Message
Date
Files
f05b06c
Merge pull request #2 from DeepaGarg-07/SCR...
7 minutes ago
1 file
dba985a
SCRUM-4: Configured GitHub Actions autom...
13 minutes ago
1 file
Issue 1 (St
Description
Add a description
Subtasks
Add subtask
Activity
Refresh

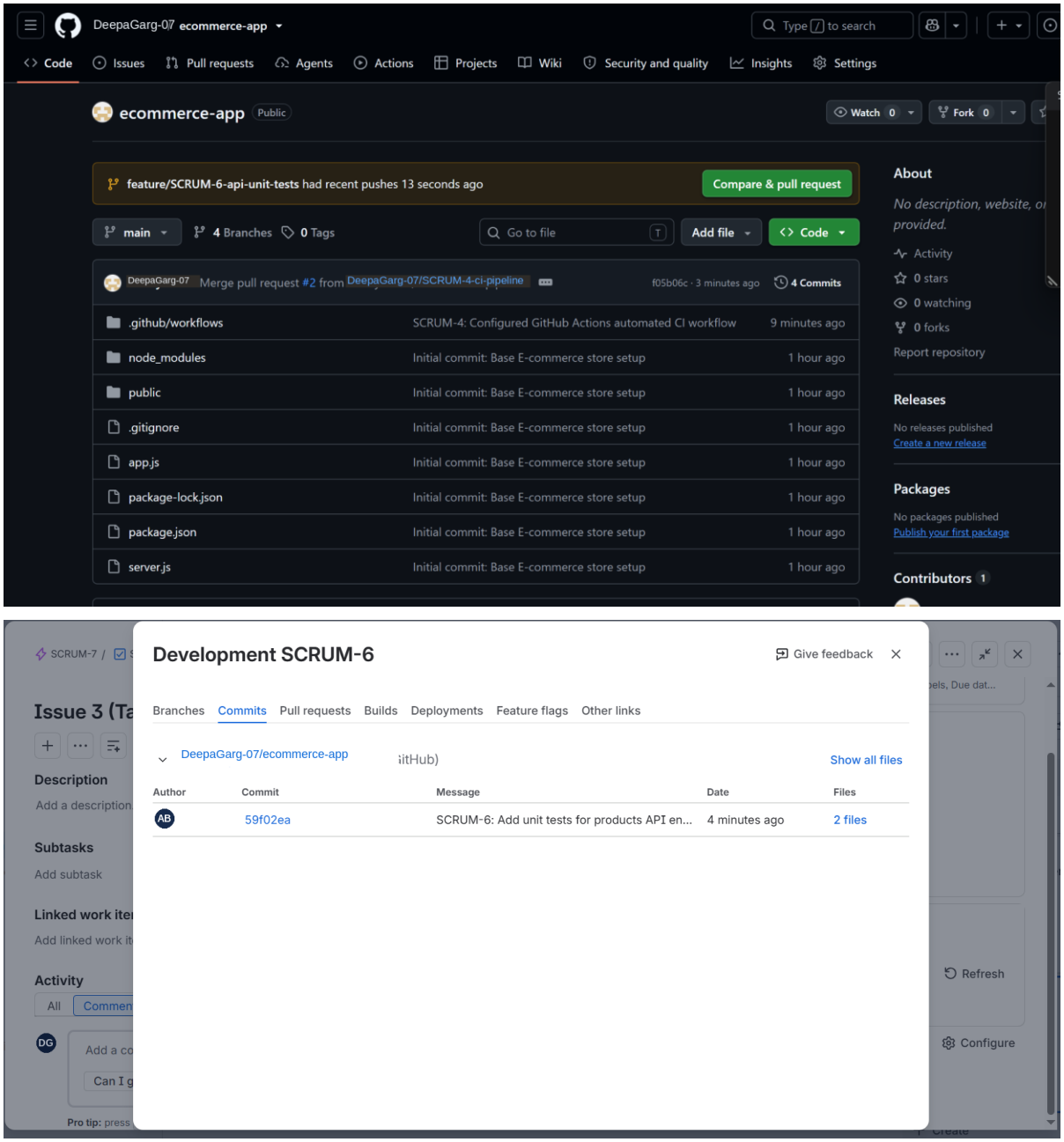

File Edit Selection View Go Run Terminal Help
Explorer
ASD-D_Deepa
ecommerce-app
node_modules
public
index.html
.gitignore
app.js
app.test.js
package-lock.json
package.json
server.js
package.json M
app.test.js
ecommerce-app > package.json > scripts > test
{
"name": "ecommerce-app",
"version": "1.0.0",
"description": "",
"main": "index.js",
Debug
"scripts": {
"test": "jest"
},
"keywords": [],
"author": "",
"license": "ISC",
"dependencies": {
"express": "^5.2.1"
},
"devDependencies": {
"jest": "^30.5.0",
"supertest": "^7.2.2"
}
}





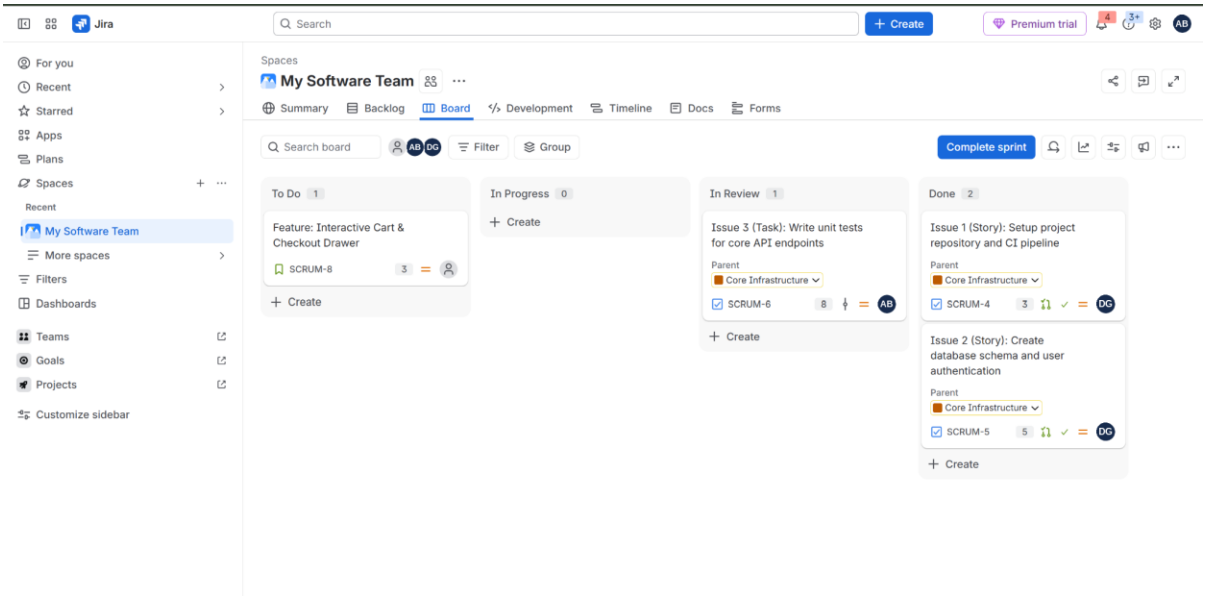
Step 10: Verifying the CI Workflow & Automated Testing

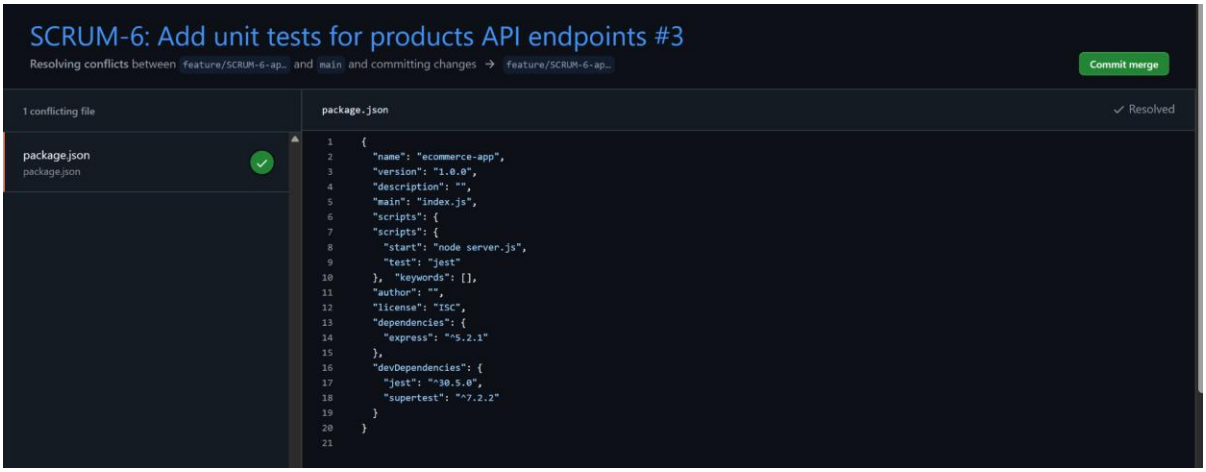
The CI/CD pipeline branch was merged into main, and the Development panel on the corresponding Jira issue confirmed the automated workflow run.

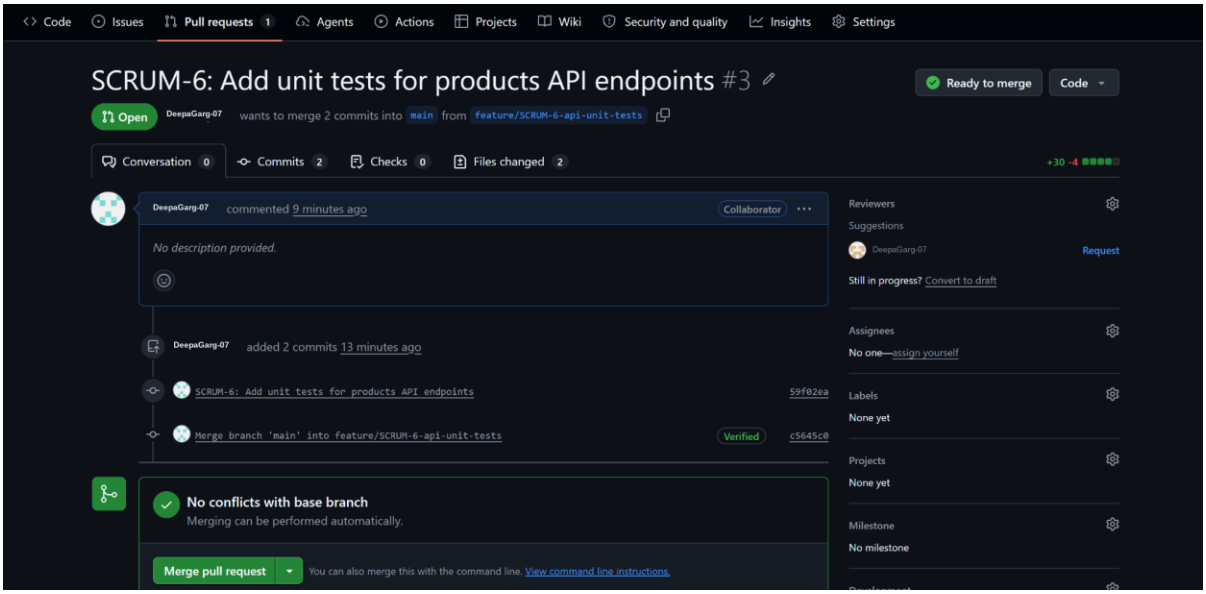


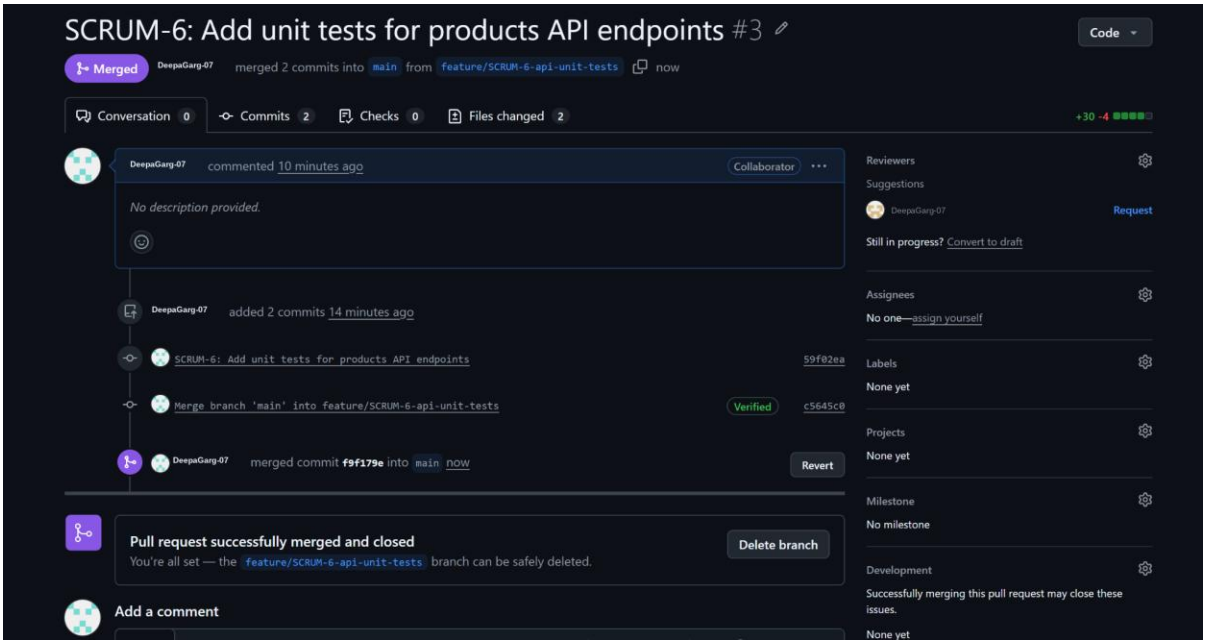
Step 11: Resolving Merge Conflicts & Finalizing Pull Requests

A second collaborator added unit tests on a separate feature branch (SCRUM-6). A merge conflict in package.json was resolved before the Pull Request was reviewed and merged into main.


Jira
Search
Create
Premium trial
For you
Recent
Starred
Apps
Plans
Spaces
Recent
My Software Team
More spaces
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
My Software Team
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Search board
Filter
Group
Complete sprint
To Do 1
Feature: Interactive Cart & Checkout Drawer
SCRUM-8
Create
In Progress 0
Create
In Review 1
Issue 3 (Task): Write unit tests for core API endpoints
Parent
Core Infrastructure
SCRUM-6
Create
Done 2
Issue 1 (Story): Setup project repository and CI pipeline
Parent
Core Infrastructure
SCRUM-4
Issue 2 (Story): Create database schema and user authentication
Parent
Core Infrastructure
SCRUM-5
Create

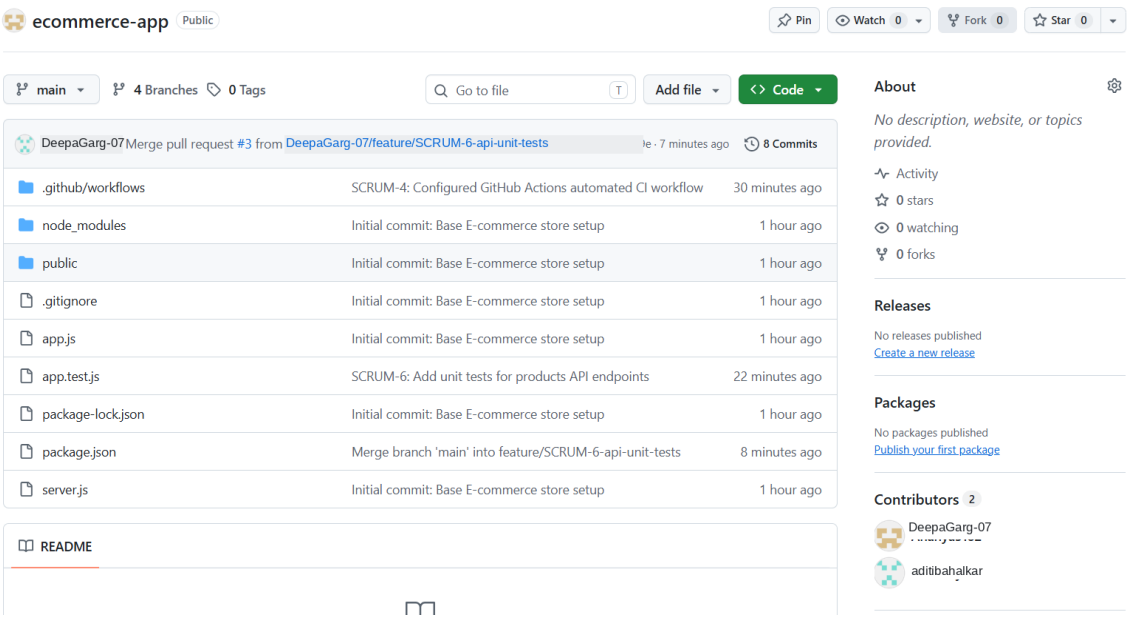

SCRUM-6: Add unit tests for products API endpoints #3
Resolving conflicts between feature/SCRUM-6-ap... and main and committing changes → feature/SCRUM-6-ap...
Commit merge
1 conflicting file
package.json
package.json
package.json
Resolved
{
"name": "ecommerce-app",
"version": "1.0.0",
"description": "",
"main": "index.js",
"scripts": {
"scripts": {
"start": "node server.js",
"test": "jest"
}, "keywords": [],
"author": "",
"license": "ISC",
"dependencies": {
"express": "^5.2.1"
},
"devDependencies": {
"jest": "^30.5.0",
"supertest": "^7.2.2"
}
}

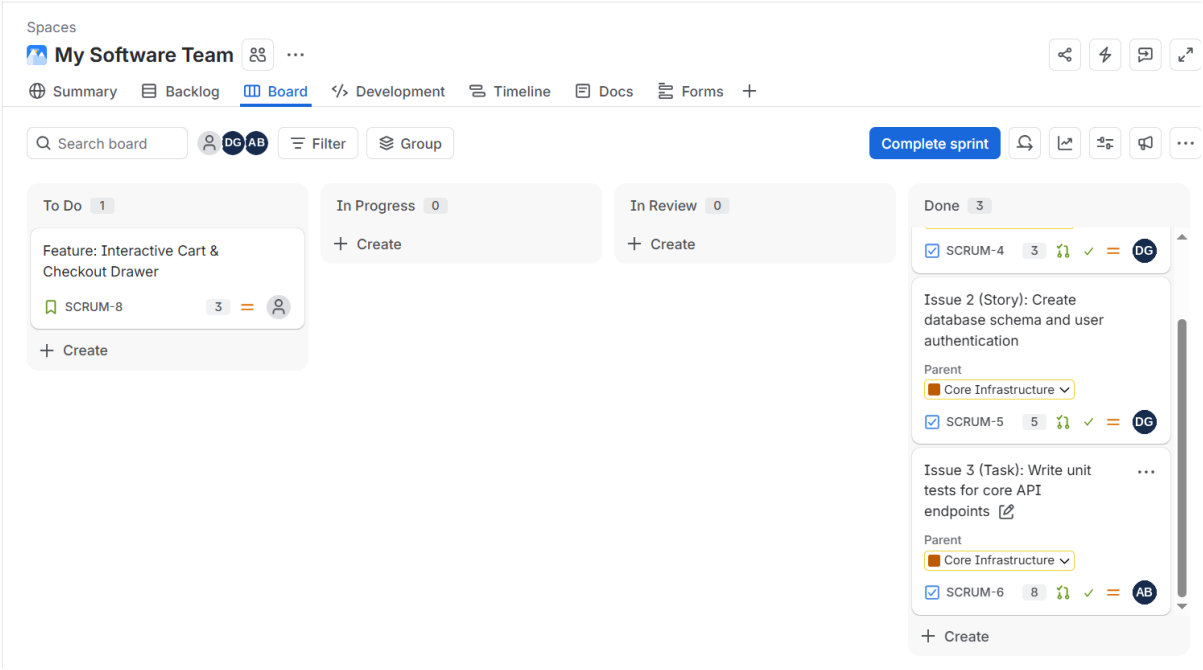

Code
Issues
Pull requests 1
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
SCRUM-6: Add unit tests for products API endpoints #3
Ready to merge
Code
Open
DeepaGarg-07
wants to merge 2 commits into main from feature/SCRUM-6-api-unit-tests
Conversation 0
Commits 2
Checks 0
Files changed 2
+30 -4
DeepaGarg-07 commented 9 minutes ago
Collaborator
No description provided.
DeepaGarg-07 added 2 commits 13 minutes ago
SCRUM-6: Add unit tests for products API endpoints
59f02ea
Merge branch 'main' into feature/SCRUM-6-api-unit-tests
Verified
c5645c0
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.
Reviewers
Suggestions
DeepaGarg-07
Request
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone

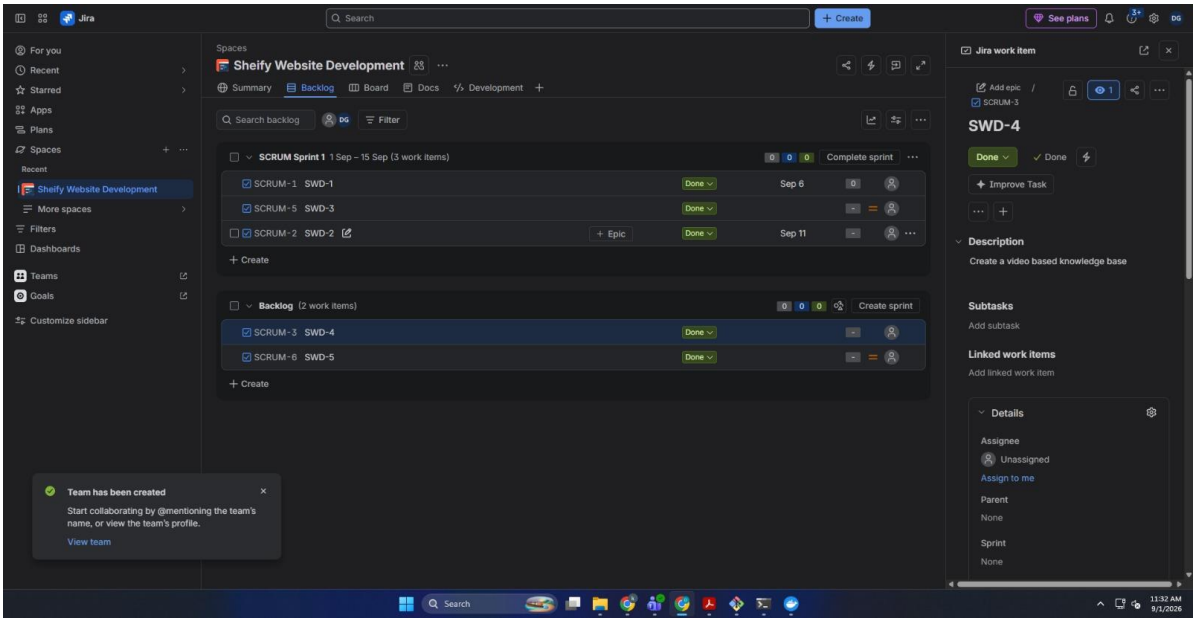


Step 12: Final Repository State & Sprint Completion

The repository reflects contributions from both team members, and the Scrum board shows all sprint items successfully moved through to Done.







Conclusion:

Integrating Jira with GitHub established an automated, end-to-end Agile development lifecycle for our E-commerce application. We learned how to organize backlogs into active sprints, assign tasks across team members, and track real-time progress using Scrum boards. By referencing issue keys in branch names and Pull Requests, we enabled seamless traceability between our codebase and Jira tickets. Furthermore, setting up GitHub Actions CI pipelines and Jira Automation demonstrated how modern DevOps workflows eliminate manual overhead by automating testing and board transitions upon code merges.